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A train rescheduling model integrating speed management during disruptions of high-speed traffic under a quasi-moving block system

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ABSTRACT

Chinese high-speed railways faced a fast development in recent years. Their performances are still confronted with disruptions unavoidably, which impact on the reliability of the traffic and passenger satisfaction. This paper presents a rescheduling model which is able to solve the critical problem of effective disruption management (namely, fast and dynamic train speed adaptation, supervision of braking and changing train sequence due to incidents, warnings or alarms), and consider in detail the signalling and safety systems based on a quasi-moving block system with variable headways. We integrate the modelling of efficient traffic management measures and the supervision of speed, braking and headway in one general job-shop model. We use a commercial solver with a custom-designed two-step method to speed up the procedure in order to solve instances from real-world high-speed networks in China quickly. Overall, the approach guarantees the resolution of the traffic control and speed management within few minutes of computation time. The output demonstrates that the proposed approach can achieve a reduction of train delays by 70% compared to the solution determined by keeping the order of the original timetable, and get the optimality for more than 90% of instances with a realistic case.

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1. Introduction

The development of high-speed railways in China has been fast and impressive. In recent 10 years (2007–2017), more than 20,000 km of high-speed rail track have been made available, which are utilized by more than 4,200 high-speed trains per day. The service is being recognized as very attractive to passengers due to the short travel time and the high speed, which may even exceed 300 km/h. However, this system is not immune to delays and disruptions caused by some inevitable events, alarms, or technical failures. Some of those effects can result in temporary interruptions of operations (see Zhan et al., 2015) or more commonly, conflicts, primary delays, or speed restrictions of trains. These generate knock-on delays and further disturbances, which reduce the benefit of high-speed trains, result in longer travel time and degrade operations reliability. These incidents decrease greatly the satisfaction and travel comfort of the passengers, and ask for increased workload of the dispatchers to adjust the railway traffic during operations.

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As mentioned in Zhan et al. (2015), the operating rules of high-speed trains are different than those of conventional speed trains, and require new approaches, in order to integrate the safety-critical elements of the operating rules. In fact, operating speeds above 200 km/h require continuously updated accurate distance headway between consecutive trains according to advanced signalling and safety systems such as moving blocks, other than the traditional 3-aspects fixed-block signalling systems, and moreover more prompt adjustment in the case of abnormal situation. This often involves dynamic train speed reductions and fast dispatching measures like changing train orders or running times. Dispatchers of high-speed railway lines are challenged to update tight train schedules and solve disruptions under strong time pressure.

Real-time rescheduling of railway traffic (which we call traffic management or traffic control) has gained an increasing recognition in recent years due to the development and deployment of intelligent traffic management systems, but there are still very few approaches, which can properly model the dynamics of high-speed railway lines. In fact, there is currently no real-time rescheduling model that can optimize the train operations with the quasi-moving block signalling system of the Chinese high-speed railway lines in the case of disruptions.

However, a prompt and reliable decision support for dispatchers in the case of disruptions requires the integration of a real-time rescheduling model at a microscopic scale and at the same time a highly accurate real-time train speed management. Furthermore, the majority of research on microscopic real-time rescheduling are confronted with small primary delays, and avoid large scale adjustments corresponding to disruptions, strong speed limitations, and blockages. Finally, among the literature dealing with disruptions, the most common situation considered is a black/white blockage, while a more realistic situation is a gray-area of speed reduction due to warnings or degraded operating conditions. Implementing speed reduction policies is crucial to keep safety and traffic going to avoid a full blockage, but reduces the transport capacity; conversely, travel times of trains will increase for a given period of time. When this is combined with the possibility of small random delays, the need to quantitatively support human dispatchers by quick and effective traffic control is evident. We therefore address the real-time problem of rescheduling high-speed railway services to reduce delays, under a quasi-moving block safety system, when the trains are disturbed by delays and disruptions affecting the operating speed. This complements the macroscopic optimization model proposed by Zhan et al. (2015) which focused on full blockages and neglected some operational principles affecting microscopic operations and speed of trains.

More precisely, the key contributions of our approach are:

1. A job-shop scheduling model on a microscopic scale is presented, describing the existing quasi-moving block signalling system in China by means of alternative graph formulation;
2. The model integrates one single optimization process with two separate decisions, i.e., traffic management (orders and times) and train control (speed management), which were typically dealt with in a decomposed, iterative or non-optimized manner in previous research;
3. The model is used for solving disruptions, which affect the operating speed of trains for a variable time and geographic extension, with a given start and end time within the optimization horizon and represent the most common issues on Chinese high-speed lines, and concurrent shorter delays perturbing further the traffic;
4. A two-step procedure is proposed to speed up the computation of upper and lower bounds, which enables a considerable improvement compared to using a commercial solver;
5. We test our model on the Wuhan-Guangzhou high-speed railway line with different degrees of speed limitation and a realistic set of primary delays;
6. A comprehensive analysis of two different operating plans (with homogeneous/heterogeneous speed traffic) identifies the potential improvement compared to several typical benchmark strategies;
7. The model can be used as decision support for train dispatching to improve the performance of high-speed railway operations in China and elsewhere.

The remainder of the paper is organized as follows. Section 2 presents a comprehensive literature review on railway real-time rescheduling, for disturbances/disruptions, and with/without consideration of train speed. In Section 3, we explain the problem, the features of the quasi-moving block signalling and safety system, and describe the typical disruptions faced. Then, a mixed integer linear programming model is formulated in Section 4. Section 5 reports the main computational experiments. A summary of conclusions and recommendations for future research is closing the paper in Section 6.

2. Literature review

The possibility to reschedule railway traffic to reduce impact of delays and disruptions has been attracting much attention in recent years. We here go quickly through the available scientific literature, clustering together mathematical optimization approaches which focus on the general problems of normal speed railway traffic; those that focus on disruption management; and those that include explicitly speed modelling and management in their analysis. We direct for more information to the recent surveys in Cacchiani et al. (2014), Corman and Meng (2014) and Fang et al. (2015).

2.1. General delay management

Several approaches based on operations research techniques are widely used to update a timetable when small delays occur, by producing a disposition schedule. This latter is the product of control actions such as retiming (change passing

time at stations), reordering (change train *order* at shared infrastructure elements) or rerouting (change *route* at stations or in the network), which aim at reducing the negative influence of delays and limiting delay propagation. A particularly popular model, the alternative graph model, uses a combination of job shop and alternative graph techniques (D'Ariano et al., 2007b); this model has been used in other industrial settings as well, referring to multiple relevant objective functions (Samà et al., 2017b). This stream of approaches model each block section as a single capacity server, which can only be utilized by at most one train at the same time, with further no-store constraints and blocking constraints relating to the processing over multiple adjacent block sections (D'Ariano et al., 2007b). The problem of rerouting trains has been considered by a variety of metaheuristics, ranging from Tabu Search (Corman et al., 2011b) to variable neighbourhood search (Samà et al., 2017a), which are able to deliver a better solution in a small computation time. Additionally, the decision support system-ROMA (Railway traffic Optimization by Means of Alternative graphs) (D'Ariano, 2009) based on the optimization library AGLIBRARY (Alternative Graph LIBRARY software) (D'Ariano et al., 2014) has been developed to compute the solutions to those classes of problems.

The addition of extra constraints to increase overall realism has been the next challenge: connection management for passenger flows with a limited microscopic detail (Dollevoet et al., 2012) or a comprehensive microscopic detail (Corman et al., 2016), consideration of multiple classes of running traffic (Corman et al., 2011a). Over all those developments, alternative graph models showed their potential to mitigate the influence of delay, decrease travel time, and include most relevant operational requirements. Complementary studies involved studying the effect of stochastic disturbances and variations for running and dwell time which might not be totally known at the moment of optimization (Larsen et al., 2014). A variety of objective functions have been studied and analysed, which include the minimization of secondary (consecutive) delay, final delay (tardiness), combined in sophisticated multi-objective optimization framework in Samà et al. (2015). The instances so far tackling with the alternative graph range from stations and corridors to multiple interconnected networks of coordinated dispatching areas (Corman et al., 2014b), but so far high speed railway traffic has been neglected.

While all the previously mentioned alternative graph approaches refer to the fixed block signalling, the model is also able to deal with moving block signalling system: Mazzarello and Ottaviani (2007) used an alternative graph model to understand implications of train control under a moving block signal environment. In this case, the headway between trains was computed as a minimum time lag on each section for two consecutive trains.

Other approaches have also been proposed to solve the same problem. Meng and Zhou (2014) proposed a different model based on cumulative flows on a time space network, which could have a variable detail approximating microscopic operations. The original complex rerouting and rescheduling problem was decomposed into a sequence of single train optimization sub-problems. In the solution process, a Lagrangian relaxation solution can provide a better lower bound. Törnquist and Persson (2007) established a mixed integer linear programming (MILP) model to provide rescheduling solution under disturbances, by the means of train scheduling and routing, for an *N*-tracked railway traffic. To improve the efficiency of the computation process, a greedy algorithm was developed to reduce the computation time (Törnquist Krasemann, 2012). Another approach also based on MILP, but exploiting Benders decomposition to provide optimal solutions, has been proposed by Lamorgese and Mannino (2015), and was applied to various networks in Norway. Pellegrini et al. (2014) formulated a MILP to tackle the real-time railway traffic management problem (rtRTMP), considering a granularity of detail including track circuits, which make it possible to demonstrate a typical fixed block signalling mechanism like the route-lock sectional-release interlocking. They studied the problem in the case of simple junctions and more complex areas, and used CPLEX to solve the model. Samà et al. (2016) further studied how to select the most promising train routes beforehand, to be included in the large and complex MILP determined by Pellegrini et al. (2014).

From the analysis above, the trend is evident for models considering microscopic detail, i.e., including signals, switches, and reporting precisely the headway between trains, to avoid risk of delivering infeasible solutions, but this comes often with an increased computational cost.

2.2. Disruption management

We consider stronger disturbances which result in a serious reduction of capacity, for instance, full or partial blockage, or serious speed restriction, as disruptions (see reference Cacchiani et al., 2014). Those require typically stronger control measures and have attracted attention only later. In the case of a blockage of a block section, the problem is then which train could use the very limited capacity left (sequencing), which train might use different routes in the network (global rerouting), which train can be cancelled (cancelling). Some approaches also consider constraints related to the circulation of crew and rolling stock in the case of cancellation or global rerouting.

An event activity network is a typical tool to solve disruptions for railway traffic with a macroscopic detail, as it allows for faster resolution and larger geographical scope. Zhan et al. (2015) used a macroscopic event activity network to reschedule the operations, when an open track of a high speed railway was totally blocked, without considering rerouting. They neglected the detailed information about routes and signalling system, and focused on the capacity of stations in terms of numbers of siding tracks in each station. A solution was found by solving the MILP by CPLEX, aiming to minimize the number of cancelled trains and total weighted delay. A whole day timetable of Beijing-Shanghai high-speed railway was treated as an experiment. Veelenturf et al. (2015) also used the event activity network by considering the infrastructure, rolling stock capacity and alternative geographical routes, when handling large-scale disruptions. Those two works focus on

the full blockage of open track caused by disruptions. There is also another research work associated with the blockage of one (more) track: Binder et al. (2017) proposed an integer linear programming formulation with three objectives to take the passenger satisfaction, the operation costs and the deviation from the undisrupted timetable into consideration during the real-time rescheduling process for timetable, from a macroscopic level.

A different stream of research (Corman et al., 2010, 2014a; Corman and D'Ariano, 2012) aims at the determination of microscopic operations, for a relatively smaller problem, once a disruption (blockage) appears. Based on an alternative graph model, a set of tentative corrective actions is determined on beforehand, and is evaluated on the basis of a range of performance indicators. The uncertainty in the disruption length was specifically considered by probabilistic approaches such as Meng and Zhou (2011). They designed sequencing actions to minimize the expected delay in a manner which was robust against the uncertainty of the disruption, for increasing traffic flow and varying disruption intensity. Later on, Meng et al. (2016) determined control actions including rerouting, which were robust against uncertain disruptions.

Not all disruptions are full blockages. Quaglietta et al. (2016) reported on a large study including many of the (alternative graph based and MILP-based) scheduling models cited so far, in order to handle disruptions which limit the speed of trains, but do not block totally the traffic. However, the disruption in this case was considered a static, totally known and fixed on beforehand, speed limitation, lasting one day. This basically resulted in changing the maximum speed of all trains running on an open track area, without further complexity from the modelling part. A similar problem (speed limitation for one open track area) has been investigated by Wang et al. (2014). In this case, the traffic was high speed, however the information on speed limitation was used to evaluate the railway capacity before the rescheduling took place, actually separating the problem of speed limitation and rescheduling.

2.3. Speed management

A third direction which has been mostly separated from the previous two, with some exceptions, deals with the control of train speed. Most of the approaches identified deal with train speed with very rough approximation, at least when optimizing. A general overview of issues related to speed management can be found in the survey (Albrecht et al., 2011).

For operations according to the schedule, there is a large corpus of research available by now, which are able to compute the regimes to be used, and optimally follow the path of minimal energy consumption, given a running time, for instance Chevrier et al. (2013), Wang et al. (2013) and Wang and Goverde (2016). Haahr et al. (2017) used a novel dynamic programming methodology for optimizing the train speed profile problem in case of unexpected changes in speed restrictions or timings. In the case of delays, the problem is the interaction of traffic. The usage of Automatic Train Operations and Communication Based Train Control (ATO and CBTC) is the most common paradigm to achieve precise control of running traffic (Albrecht et al., 2011). For instance, Domínguez et al. (2012) focused on the design of optimized ATO speed profiles. The application of such approaches to non-metro lines had anyway strong limitations in the variation possible for the delays (only minor adjustments possible, no reordering), and amount of trains considered. The approach implemented in the Lötschberg tunnel was described in Montigel (2009), that simulated only very few trains at a time, in terms of train and traffic characteristics. The approach was reputed with very good performances, but it was limited to a well-defined small test case, with a limited traffic. The optimal solution could be found by exhaustive search, but it suffered from scalability and applicability to larger networks. The innovative approach proposed by Rao et al. (2013) aimed at pushing this concept further, but this was not yet quantified or finalized. It might involve some heuristic extension of the previous work, addressing open issues on scalability and applicability to general networks and heterogeneous traffic.

In the general case of traffic delayed and rescheduled, the most common approach is the sequential adjustment of the speed profiles, based on a scheduling solution which approximates or neglects the speed management problem (see D'Ariano et al., 2007a). In this stream, D'Ariano and Albrecht (2006) and later Albrecht (2009) focused on optimizing the energy minimization problem to deliver a continuous speed profile, given a schedule. In a different research stream, the time windows at stations and relevant points are used to give enough room in the rescheduling problem to calculate the energy-efficient speed profiles of trains (Albrecht et al., 2013). The result is optimal for energy efficiency, given the solution to the scheduling part, i.e., the passing times of trains at stations and relevant points.

A final stream of approaches includes iterative approaches which feed an optimized speed trajectory back to the scheduling model to improve traffic performance. In general, those approaches offer no guarantee of optimality in either scheduling traffic or speed management. Such approaches include the method of Mazzarello and Ottaviani (2007) for the EU project Combine (traffic and speed are both determined by heuristics, double feedback loop architecture), already mentioned. A similar approach was proposed later by Lüthi (2009). The positive feature of such approaches is that the feedback loops keep the deviation small. Having the two models separated means a match between the objectives of the two models has to be found, typically, this is the extra delay introduced by speed management. Moreover, stability, convergence, quality of the system under a closed loop feedback system are even more difficult to quantify than a corresponding sequential one.

A radically different approach to achieve minimal-energy operations is to invert the hierarchy of the problems, and firstly solve the problem of generating energy efficient speed profiles, and then use only those in the traffic management part. This has been operationally translated into a choice of speed profiles from a finite set: a single speed profile in the case of Corman et al. (2009), apart from retiming actions; multiple speed profiles in the case of Caimi et al. (2012), including retiming. Then those profiles were included in the optimization problem.

2.4. Knowledge gap

Few implementations of optimization in real-time railway traffic control have been seen in reality, see the pilots or tests reported in Mazzarello and Ottaviani (2007), Mannino and Mascis (2009), Lamorgese and Mannino (2015) and Montigel (2009). Apart from the first one mentioned, the other mathematical models (and also those that have not been implemented) deal with fixed block operations, which is the standard for conventional speed trains. While mathematical models have been designed for rescheduling under the new moving block technology in Mazzarello and Ottaviani (2007), no practical rescheduling implementation of moving block technology is available currently. Quasi-moving block technology is used in an impressively growing network, especially in the largest developing market for high speed traffic in China. The main reasons for this usage are its capability of accurate separation for trains and high utilization of railway capacity, as this signalling system makes it possible to enforce a variable safety distance in fixed block infrastructure. On the other hand, no mathematical model is actually available for that case. This paper aims to bridge this gap.

A first knowledge gap arises from the slight shift in meaning due to the signalling system. Conventional (fixed block) train operations require separation over fixed and static block sections, one section can only be utilized by at most one train at the same time. This is a space headway, which requires a microscopic model to be properly reported. Macroscopic models cannot represent precisely this space headway, as the signal aspects are not considered, but instead refer to a simplified time headway. This results in further approximations neglecting the physical infrastructure capacity of the networks. For modelling the safety system of high-speed operations, a microscopic model is needed; but the headway should be considered as a variable space/distance, depending on the speed.

This brings about a second major knowledge gap, resulting from the interconnection of traffic control and speed management, required to represent the functioning of the signalling system. To this date, no approach is known which is able to optimize together speed and traffic in the case of high-speed traffic or disruption management. The majority of the train rescheduling approaches currently separate those two problems, considering either fixed speed profiles only, or oversimplifying the scheduling problem. The speed of trains determines the variable space/distance headway between trains. In real operations and especially under disruptions, we cannot assume all traffic going in free flow (undisturbed) conditions, thus a model has to be able to consider variable minimum running times, according to acceleration or deceleration features. Instead, the vast majority of the previous works consider a constant, pre-determined, static minimum running time, based on undisturbed operations of the timetable and lacking any interaction between speed and management.

Finally, the current research on disruptions mostly focuses on full or partial blockages, and avoids speed limitations, probably due to the very simple reason that most rescheduling models neglect speed values. Although speed limitations are considered, the speed limitations are set to affect the entire open track section between two consecutive stations, for the all duration of traffic control, which allows to determine the minimum running time of trains on the impacted area independently (on beforehand) from the scheduling solution. This kind of speed limitation can be easily dealt with, if the traffic has a homogeneous speed. In the proposed model, we refer to a variable range of speed limitation on a set of several block sections in an open track, with a given start time and end time both falling within the optimization horizon. Combined with the presence of a primary delay (deviation of entry time), the minimum running times for those potential impacted sections are unknown on beforehand for each train, instead they are dependent on the scheduling solution. This flexible speed limitation will lead to more conflicts, since a heterogeneous running time is considered in our rescheduling process. Moreover, none of disruptions is the same: different geographical areas are impacted, with varying start time and end time, and maximum permitted speed value, which are related to the ever-changing phenomena seen in real life, like failure of one track, heavy rain, alarms, etc. Those kinds of disruptions are very common in Chinese high speed railway system, which is explained in Section 3.2 in detail.

To address those gaps, we integrate the traffic control and speed management to deal with flexible disruptions related to speed limitation for the Chinese high-speed railway. The real-time speed of trains, their relative distance headway and a speed possibly depending on the current disruption are all considered in this paper, thus a developed microscopic model based on the space headway realizes the dynamic adjustment between train speed and safety distance. Meanwhile, the minimum running time depending on the speed level of each train on each section is determined during the solving process, which is a novelty compared to the current research. Additionally, the method can connect the traffic transition from a normal condition to a disruption condition smoothly and recover to the normal traffic. This paper is thus a complement to the stream of studies improving management of high speed traffic under full blockage disruptions (Zhan et al., 2015), but requiring much more detail in the level of rescheduling problem (microscopic detail) and speed management (i.e., coupling scheduling and speed management).

3. Conceptual illustration and problem description

3.1. Chinese train control system-3 (Quasi-moving block system)

Most of Chinese new-built high-speed railways use Chinese Train Control System 3 (CTCS-3) to maintain operational safety of movements of signalling and interlocking system. CTCS-3 is similar in some aspects to ETCS level 2, apart from the fact that it is a quasi-moving block system. In CTCS-3 (see Fig. 1), Centralized Traffic Control (CTC) can send information and/or commands to control the operation of switches, signals, track circuits based on the real-time train status, through RBC (Radio Block Center), CIC (Centralized Interlocking Centre), TCC (Traffic Control Center) and so on. At the same time,

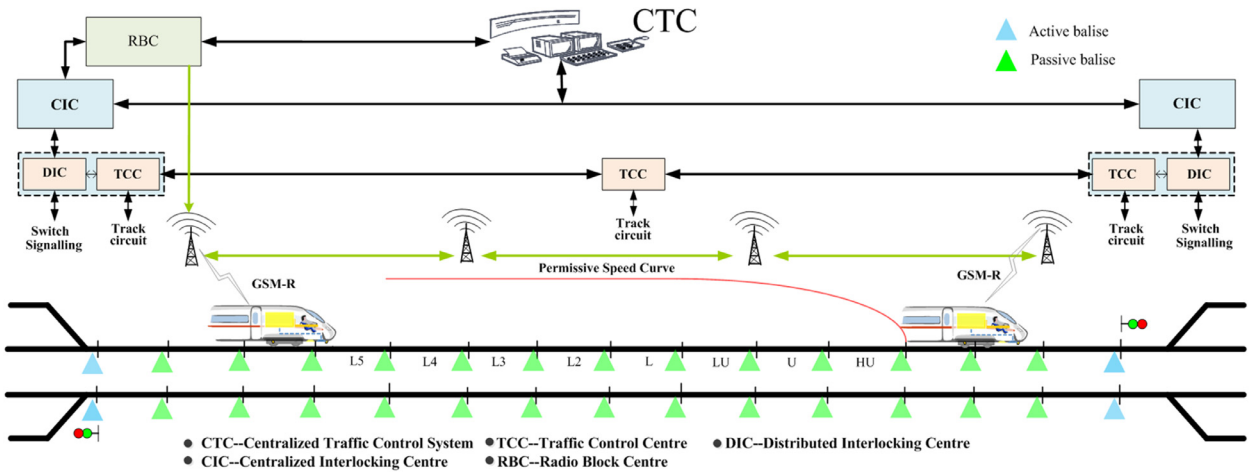


Fig. 1. Illustration of CTCS-3. RBC = Radio Block Centre; CIC = Centralized Interlocking Centre; CTC = Centralized Traffic Control System; TCC = Traffic Control Centre; DIC = Distributed Interlocking Centre (Guo et al., 2008).

the RBC can also send movement authorities, speed advices and information related to speed limitation to on-board equipment by GSM-R. On the one hand, the real-time location and speed of trains is transmitted to the RBC by GSM-R radio communication. On the other hand, fixed equipment such as active or passive balises can also send fixed parameters of the tracks, and possible dynamic speed limitation to the on-board system. The track circuit can sense the occupation of track sections ahead, and send the amount of free sections ahead of the train, for computing movement authority. Then the on-board system generates a permissive speed curve according to what the information available is, both from the RBC and the balises. The permissive speed curve matching the movement authority ends at the beginning of the section utilized by the proceeding train (i.e., fixed block), instead of the tail of the former train (as it would be common in a moving block). Thus while position is continuously monitored, and movement authority is continuous, the track detection system requires a given amount of (fixed-block) track sections ahead to be free. We refer to this train operation as quasi-moving block safety and signalling system. More details about CTCS-3 can be found in Guo et al. (2008). Overall, this very advanced signalling system offers the great potential for delay reduction, similar to ETCS (see Goverde et al., 2013). Currently, ETCS-level 2 systems are operated in Switzerland, Italy, the Netherlands, Germany, France, Sweden, and Belgium on high-speed railway lines.

In a very short summary, all signalling systems work with protected zones, where the braking distance of a train (protected zone) has to be kept free from other trains. In the fixed block signaling system, the protected zone is a fixed amount, and at fixed locations, corresponding to the physical block sections. In the moving block signaling system, this is a variable amount related to speed, and moving, but it is unrelated to infrastructure features. In quasi-moving block signalling, the protected zone is variable related to speed, but corresponding to physical block sections. In other terms, in a moving block signalling system the block sections are virtual; a moving blocking system and the safety distance in front of a train must end before the real location of the tail of the preceding train. In a quasi-moving blocking system, the safety distance in front of a train must end before the beginning of the block section occupied by the preceding train. The computation of speed profile is similar in both kinds of signalling systems. For a longer discussion on (traditional) fixed block signalling systems, on advanced moving block systems such as ETCS-level 3 and on intermediate solutions like the quasi-moving block systems, we refer to Hansen and Pachl (2014).

More precisely, the permissive speed curve considers the actual speed, number of free sections, and is further dependent on the maximum speed of trains and the condition of tracks. In fact, as the speed grows higher, the running time over the sections decreases, meanwhile the braking distance increases. This asks for an amount of free block sections which is larger than one (as it is common in three-aspects signalling system): the amount of sections to be free between two successive trains mainly depends on the speed of the second. If the required amount of free block sections in front of the train is not available, the speed of the train must be reduced to a lower level, compatible with the movement authority, and the running time will correspondingly increase. A schematic representation of the requirements on free block sections (B), based on the speed (v) is given here below in Eq. (1). For instance, when going at 300 km/h, at least 5 block sections should be free ahead of the train. When the speed is below 120 km/h, the system behaves as a traditional three-aspect fixed block signalling system.

$$B = \begin{cases} 5 \text{ if } 250 < v \leq 300 \text{ km/h} \\ 4 \text{ if } 200 < v \leq 250 \text{ km/h} \\ 3 \text{ if } 160 < v \leq 200 \text{ km/h} \\ 2 \text{ if } 120 < v \leq 160 \text{ km/h} \\ 1 \text{ if } v \leq 120 \text{ km/h} \end{cases} \quad (1)$$

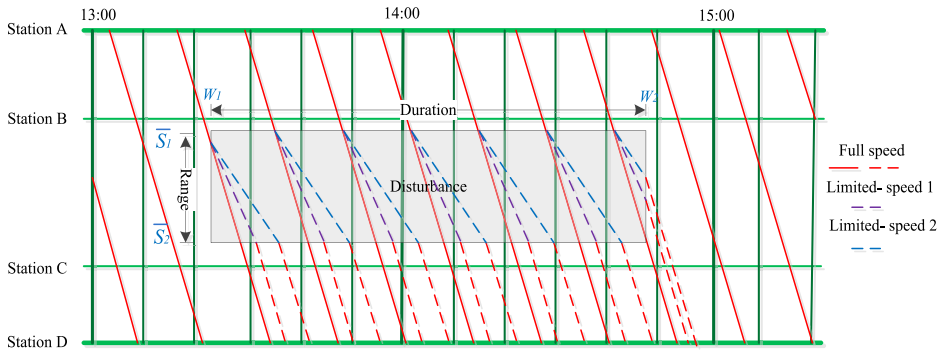


Fig. 2. Parameters of disturbances related to speed limitation. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

3.2. Disruptions considered

The range of disruptions in railway systems can extend from primary delays due to external processes (boarding and alighting), running time extensions caused by speed limitation, planned maintenance for checking or fixing problems, to abnormal behaviour of systems (related to safety braking, failures of subsystems, accidents and collisions). In Chinese high-speed railway operations, many kinds of alarms can trigger speed limitations measures, such as bad weather, failures of the infrastructure or electric system, as well as maintenance requirements; those are quickly discussed in what follows.

Bad weather refers to strong wind, heavy rain, snow, flood, hurricane, mud-rock flow, earthquake, etc. When an alarm for bad weather is triggered in real-time by some sensors along the railway line, the dispatcher usually issues a speed limitation command for the geographical area influenced, according to the severity degree of the situation. In general, this disruption impacts both tracks of a high-speed railway network at the same time, and in the same manner. A failure of the track system or the electric system might relate to a bad alignment of the track, and/or the electric system. When such a situation is detected, a speed limitation is issued for the concerned track and/or the opposite track. With regard to the disruptions related to bad weather, their duration and severity degree are generally uncertain, but the current climate forecasting technology is able to provide the relevant approximate parameters. In contrast to those (uncertain) disruptions, maintenance activities are more predictable, and have the precise duration, geographical extent and severity set beforehand. In general, to ensure the safety of maintenance staff and railway operation, low running speed is required during maintenance, or just after them (when the areas being maintained are still being checked).

According to the recorded disruptions and alarms in a high-speed corridor in China, the incidents mentioned above can take up to 85% among all the events (Xu et al., 2016). Disruptions related to speed limitation are thus very common in high-speed railway operation system, causing large primary delays with high probability, but cancellation of trains with extremely low probability. This justifies our choice of focusing only on the adjustment of traffic with the management of speed under normal traffic or speed limitations, instead of cancellation of trains. In Chinese high-speed railways, full blockages do not normally trigger adjustments of rolling stock and crew schedule, and those latter issues can be ignored when modelling the operating rules (see Zhan et al., 2015, for details).

3.3. Problem description

This section focuses on the description and discussion of the main parameters representing the disruptions and their severity. The most important parameters relate to time (start time, end time and time span), influence range (amount of block sections affected), limited speed value (maximum allowed speed). Fig. 2 shows a time-distance plot of some trains affected by a speed limitation. The horizontal axis represents time, increasing rightwards; the vertical axis represents space, with trains going in the downward direction. The grey rectangle is the portion of time/space affected by the speed limitation: the disruption occurs between station B and station C from section $\overline{S_1}$ to $\overline{S_2}$ and it starts at $W_1$and ends at W_2, causing 7 trains delayed. Trains are depicted in red when running at full speed, and in purple and blue when the speed is limited to lower speeds (blue corresponding to the lowest speed allowed).

The characteristics of the disruptions might be described as stochastic variables, but (as common) we suppose their parameters are fully determined, once the disruption is known, and in any case when the optimization algorithm is launched. We consider the variability of possible disruptions by means of a set of relevant scenarios. Our disruptions are limited in time, and we consider operations for a certain time before and after the disruptions. We thus implicitly deal with the relevant problem of the transition periods, and determine for each train whether it will run in disrupted conditions, or not. The problem is to find the best train scheme, by reordering and retiming, as well as provide detailed speed selection for each train on each section, which together reduce the overall delay and improve the service quality.

The final result is a decision support system which decreases the workload of dispatchers and represents a recommendation for train drivers (in terms of expected speed profile) while at the same time allows for a better passenger service quality.

4. Model formulation

4.1. Job shop scheduling model

Railway lines are practically divided into block sections, which are used to ensure a separation between trains over the same infrastructure resource (section, switch, platform, etc.). We model railway traffic as a *job shop scheduling* problem, which associates with trains running (following a particular speed profile) over block sections with *operations,* and treats block sections as *resources* with unitary capacity. Formally, a train movement translates to a sequence of operations on the successive block sections of the route assigned for the train (assumed to be given). Each operation is associated with a *duration* (the running time, based on a particular speed profile), which equals the difference between the *start time* of the current operation (the time the head of the train enters the block section) and the start time of the operation on the successive block section. Most job shop models consider the duration of the operations fixed a priori; in our model, it will be a variable. Some operations also have associated with a minimum *departure time* (the entrance in the network or the departure from a station), a planned *arrival time* (at a station or at the exit of the network). Moreover, operations of different trains on the same resource / block section need to be separated by a suitable *time headway*, which might depend on the speed profile of the train, and the occupation of block sections ahead. The procedure to compute the time headway is one key feature which is peculiar to quasi-moving block operations. No train is allowed to enter a block section which is already occupied. For more details on job shop models for railway operations, we refer to D'Ariano et al. (2007b). While there is no general agreement in the literature about the objective function to use, a consistent stream of research has been using the maximum consecutive delay.

Moreover, we represent the job shop model with an *alternative graph* formulation, which has been proved to be flexible enough to include many relevant constraints. This translates the problem to a graph-based structure, where nodes are associated to block sections in the train route, and fixed arcs (solid arcs in a graph) between nodes refer to the train moving between two consecutive block sections, with a suitable weight. More formally, the weight of the fixed arc corresponds to the minimum running time for the train moving between two nodes. Determining the order between two trains on the same block section or any one section among the conflicting sections (e.g., the sections in one switch area) can be modelled by choosing one arc out of a pair of alternative arcs (dashed arcs in a graph), which are associated to the two possible orders. The weight of the alternative arc relates to the minimum headway time required between two trains, ensuring a minimum train separation time, which is greater than the minimum safety requirement. In a feasible solution, exactly one arc is selected from each alternative pair, and there is no cycle in the solution graph. In this case, the start times of all operations can be easily computed by a longest-path calculation, from a common reference point.

Compared to the basic alternative graph model introduced in D'Ariano et al. (2007b), we consider the following relevant extensions: there is not a *single* alternative pair between two trains on one block section in open track area, but a *set* of alternative pairs, with each pair of alternative arcs relating to a particular case of speed level and number of free sections required. In a feasible solution, only one arc of a single pair out of the set of pairs is selected, i.e., the choice of which pair out of the set relates to the speed levels, and the choice of which arc out of the pair relates to the order of trains. Similarly, there is not a single (predetermined) weight representing the running time for an operation, but a set of weights representing possible running times at different speeds. Again, only one of them will be selected, corresponding to the actual speed level followed by the train on that block section. To summarize, the concept of alternative arcs is extended to represent three aspects at once: firstly, the determination of the order of trains; secondly, the choice/update of the speed of the train according to the available free sections in front of the train, in terms of running time; and finally, the implications of the chosen speed towards headway with other trains. To give a pictorial description, we report in Fig. 3, an illustrative case of two trains running along a railway line, for their speed and operations modelled in the proposed extension of the alternative graph model.

Fig. 3 refers to two trains crossing a line, from left to right, named train t_1 and train t_2. Train t_1 is planned to stop at a station in the network. In this case, train t_1 enters this network with a minor primary delay, while train t_2 runs punctually. Fig. 3 has 5 associated layers, from top to bottom: the speed profiles of train t_1 and t_2; the track layout with block sections from 1 to 11, and the station on block sections 2–5; the alternative graph model; a feasible solution to the alternative graph model; and the representation of that solution in a time space graph.

The x-axis is common to all layers, and extends along the track layout; nodes in Layer 3 and 4 are reported at the location corresponding to the block section they refer to. The routes of the trains t_1 and t_2 are respectively going over the following block sections: 1-2-3(stop)–5-6-7-8-9-10-11 and 1-2-4-5-6-7-8-9-10-11.

Each row of nodes in the alternative graph model of Layer 3 represents the route associated with a train (top t_1, bottom t_2), identified by the block sections where the operations take place. Similar to the typical alternative graph, the fixed arcs (solid arcs in Layer 3) illustrate the geographical connection of block sections or requirement of technical activity. The weight of the solid arcs denotes the requirements of operation time for trains passing through the start node (previous section) of the arc. To be specific, the weight of the red solid arc relates to the entrance time, and the green solid arc represents the earliest departure time from a station. However, the weight of the dark blue arcs is a kind of running time depending on the speed level of train, which is variable, instead of a constant value in the typical alternative graph. In order to illustrate the working of the alternative arcs, we report two kinds of alternative arcs: a pair of alternative arcs, a set of pairs of alternative arcs just for the block section labelled 6, for the sake of clarity. The other sections (section 7 to section 11) will

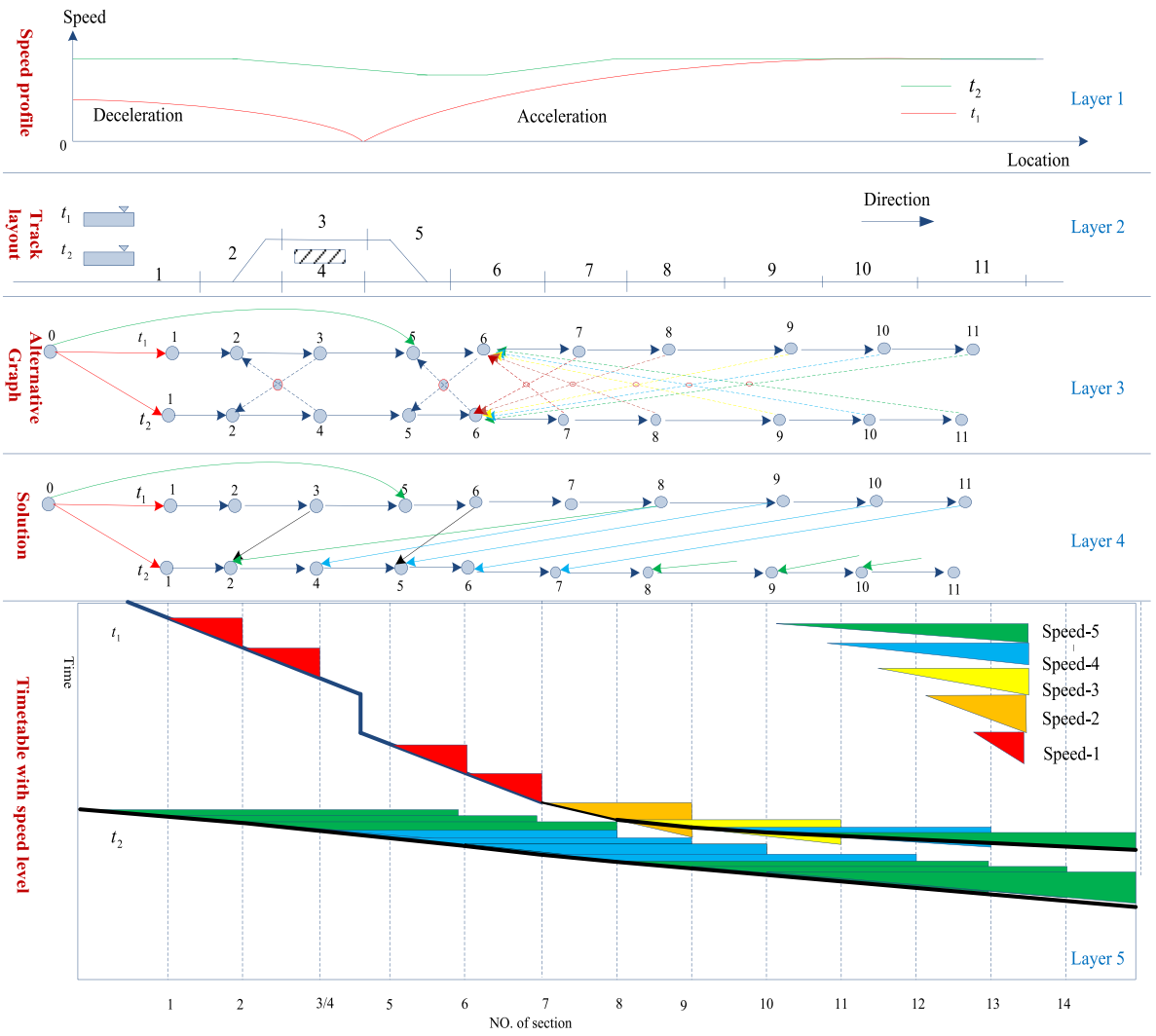


Fig. 3. Alternative graph based model of quasi-moving block train operations. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

have a similar set of pairs of alternative arcs. Alternative arcs related to node 6 in the set are reported as dotted arcs of different colours; arcs in the same pair with same colour are coupled by small red circles. The colours of the alternative arcs represent the different requirements of distance headway, related to different speed levels defined in the quasi-moving block system. When a train enters a given section with a certain speed, there must be enough free sections ahead in relation to its speed. The larger set of free block sections required becomes an alternative arc starting "more to the right" in the graph, and spanning a larger amount of block sections. The red/orange/yellow/blue/green dashed lines denote 1/ 2/ 3/ 4/ 5 available sections (minimum distance safety) required by speed level 1/ 2/ 3/ 4/ 5 (reported as B in Eq. (1)). We focus on the alterative arcs entering the nodes associated with block section 6, for train t_1 and t_2 in Fig. 3. We can identify a succession of arcs, red, orange, yellow, blue and green, which are starting from a node increasingly at the right, i.e., increasingly far away. As a result, there are 5 pairs, with 10 alternative arcs between t_1 and t_2, concerning block section 6 only. Out of those alternative arcs, only one pair (related to one colour and one speed level) is active, depending on the speed; and a single arc in this active pair has to be selected. Such a larger set of alternative arcs exists for all block sections where a high speed can be reached, typically the open track sections between two stations. On the other hand, several pairs of "normal" alternative arcs, i.e., same with the alternative arcs in the typical alternative graph, where the speed is not of importance are reported by the dash black arcs with one circle in Fig. 3, whose weight is the headway time between two consecutive trains.

The Layer 4 in Fig. 3 reports a solution to the problem, which is again an alternative graph, but where only the selected arcs of active alternative pairs are plotted. Note that the train t_2 runs after the train t_1. As t_1 precedes the other on sections 2 and 5, the arc between node 3(t_1) to node 2(t_2), and the arc from section 6(t_1) to section 5(t_2) are selected (we describe a node by its block section label, and train). The selection for both arcs determines the order of trains. In terms of alternative

Table 1
Basic parameters of railway system.

Symbol	Description
T	Set of trains
t, i, j	Index of trains
S	Set of sections;$s, p, q, h, k \in S$, index of sections
ϑ_η	Set of the η-th open track sections, where multiple speed levels are possible and trains cannot overtake,f_η is the first section of ϑ_η.
sw	Set of sections with switches.

arcs on open track, the speed of train t_2 has to be reduced to a lower speed, as there are not enough free sections to go full speed. In fact, the selected alternative arcs in section 4,5,6,7 are those coloured in blue, which means the speed level is 4. Concerning the larger set of arcs where a higher speed can be reached, there are arcs with the same colour (green) on nodes from section 8 to 11, with the meaning that the speeds associated at those block sections are level 5, the fastest.

The bottom Layer 5 of Fig. 3 reports a time-space graph of the solution identified in the alternative graph of Layer 4. Space is along the x-axis, and time on the y-axis, increasing downwards. The two trains are reported, and the colour of the triangle associated represents their speed. One can thus identify t_2 as there is no dwelling on the station, where the colours of triangles (approach and blocking space of a train) turn green (level 5) to blue (level 4) while decreasing speed, then turn back to green while reaching full speed (level 5). Train t_1 is instead reported as a steeper line, with red blocks (slower speed), a stop, and then an acceleration phase, where the colour of the blocking space shifts from red to orange, yellow, blue and reaching green (full speed) at the right end of the plot. For ease of viewing, we report the blocking space/time with different horizontal widths: a "green" blocking space represents a reservation of 5 consecutive sections, and spans thus 5 sections, while a "red" blocking space is associated with a reservation of a single section, and is thus shortest.

In high speed railway, there are typically double tracks, one for each direction, with little or no interaction between the two directions. We furthermore consider the following assumptions:

(1) No trains are cancelled under the influence of disruption-related speed limitation.
(2) The rescheduling for the rolling stock plan is neglected.
(3) The speed profile of each train obeys any limitation on maximum speed due to the safety system, and any disruption. If the train is required to slow down or even stop for any reason, the speed should follow a realistic deceleration profile allowing this deviation from the plan.
(4) The minimum running time of each section for each train depends on the speed profile selected.
(5) The route of each train is fixed. Only the main track in stations can be used for passing-through trains without boarding and alighting activities; otherwise, trains have to dwell at siding track(s) in station(s).
(6) The disruption is known at the moment of optimization, for its characteristics: the maximum allowed speed, geographical extent (block sections for which it holds), start and total duration. Further research on probabilistic approaches considering uncertainty for those values (Meng and Zhou, 2011), or to determine those values with very little error in real-time (Kecman and Corman, 2016) is out of the scope of this paper.
(7) The deceleration and acceleration performance of all trains are given.
(8) The granularity of time is at the level of seconds: durations of operations are identified within a one-second precision.

4.2. MILP model

We now translate the alternative graph model introduced in Section 4.1 to a MILP model, which can be then solved by any commercial solver. The relative definitions and parameters of railway system in this model are listed as follows. Tables 1 and 2 summarize the main parameters of train operations, and Table 3 shows the main variables used.

As reported in Table 3, there are five distinct types of decision variables; four of them concern the operations of trains, namely the operation times, orders at stations and in open tracks, and train speed. The fifth decision variable is used to determine whether a train is subject to the disruption (which has a given start/end time) or not.

Let $\omega_{t,s}$ denote the start time of train t for occupying section s. This variable can be considered continuous (real). If all durations have a granularity of second, the resolution of this variable also will be in the range of seconds.

For clarity of explanation, we separate the alternative arcs into two groups, i.e., one group in stations where there is no need to include different speeds (typically at stations, before/after a stop, as well as at the first/last section of the area), another group on open tracks, where the order of trains is fixed over a set of block sections, and where a choice of speed is available. The decisions regarding both groups of alternative arcs are modelled by binary variables. The former decision (whether train i has to enter section p after train j has entered section q) is identified as $\lambda_{i,j,\,p,\,q}$. Formally,

$$\lambda_{i,j,p,q} = \begin{cases} 1 & \text{train } i \text{ uses arc } (k,p) \text{ before train } j \text{ uses arc } (q,h) \\ 0 & \text{train } j \text{ uses arc } (q,h) \text{ before train } i \text{ uses arc } (k,p). \end{cases}$$

We report the latter decision as $\mu_{i,j,\,\vartheta}$, identifying whether train i precedes train j on all sections of the open track ϑ_η. As overtaking on an open track is not possible (by definition), the sequence of trains in the same open track ϑ_η is the same for

Table 2
Parameters of train operation.

Symbol	Description
δ_s	Operation time or running time on section s at the maximum allowed speed
$\delta_{p,q}$	Time headway of alternative arc (p, q)
R_t	Set of sections in the route of train t, $R_t \subset S$
F	Set of fixed arcs, $F=\{(p, q)\|p, q \in S\}$, p, q are two consecutive block sections along train routes
A	Set of alternative arcs, $A = \{(((j, p), (i, q)), ((i, h), (j, k)))\|p, q, h, k \in S \& (p = q \text{or} h = k)\}$
B	Set of possible speed level $\xi(\xi \in B)$, with a small abuse of notation, B also denotes the amount of free sections required between two consecutive trains by signalling system.
$\overline{B_m}$	Maximum allowed speed level
B_m	Maximum allowed speed level under a disruption-related speed limitation, $B_m \in B, B_m < \overline{B_m}$
Δt_ξ	Additional running time (compared to full speed) when train runs at speed level ξ
D_t	Set of (siding) block sections at stations used by train $t$to dwell, $D_t \subset S$
$e_{t,s}$	Time at which train t is entering on its first section s (entrance time)
E_T	Set of planned entrance sections in the controlled area for all the trains, $E_T=\{t \in T, s \in R_t, s$ is the first section of $R_t\}$
AD_t	Set of the first, last section in the route of train t, or the last/first section before/after a planned stop, whose the maximum speed level $B_m=1$
$d_{t,s}$	Dwell time of train $t$at station, regarding block section s, $s \in D_t$
$p_{t,s+1}$	Planned departure time of train t at the first section $s+1$ after station $s \in D_t$
dt_t	Primary delay time of train t
c_t	Planned total travel time of train t
ω_0	Reference start time of dummy origin node, here $\omega_0=0$
$\bar{S}$	Sections with disruption along a corridor, $\bar{S} \subseteq S$, $\bar{s}_1$ ($\bar{s}_2$) is the start (end) section
$[W_1,W_2]$	Time window of disruption,W_1,W_2 represent the start and end time respectively.
M	A sufficiently large positive number

Table 3
Variables description.

Symbol	Description
$\omega_{t,s}$	Start time, indicating the moment when train t begins to occupy section s
$\lambda_{i,j,\ p,\ q}$	Alternative arc choice: indicating the order of train i, j on block sections p and q
$\mu_{i,j,\ \vartheta}$	Alternative arc choice: indicating the order of train i, j in open track ϑ
$\beta_{\xi,t,\ s}$	Speed level choice: indicating the speed level ξ of train t on section s
$x_{t,s}$	Disruption effect: indicating whether $\omega_{t,s}$ occurs before the start time W_1 of disruption or not
$\bar{x}_{t,s}$	Disruption effect: indicating whether $\omega_{t,s}$ occurs after the end time W_2 of disruption or not

any section therein, thus that can be translated into a relation between start times of any operation on the open track. As a result, we restrict the orders to the first section f_η of an open track ϑ_η only, to assess the sequence of trains. Formally,

$$\mu_{i,j,\vartheta_\eta} = \begin{cases} 1 & \omega_{j,f_\eta} \geq \omega_{i,f_\eta} \\ 0 & \text{otherwise}, \end{cases}$$

The headway between two trains on each section in an open track depends on their speed, which is described by the speed level ξ, out of the discrete set B. We identify a set of binary variables $\beta_{\xi,t,\ s}$, which determine the speed level ξ for train t on section s. Formally,

$$\beta_{\xi,t,s} = \begin{cases} 1 & \text{if train } t \text{ runs on section } s \text{ at speed level } \xi, \\ 0 & \text{otherwise}; \end{cases}$$

Finally, a binary variable is introduced to model whether train t is impacted by the speed limitation disruption in section s, i.e., whether it will run on the time and geographical place of the disruption or not. We use a combination of two binary variables $x_{t,s}$ and $\bar{x}_{t,s}$, which determine whether train t is running before or after the disruption (which exists between time W_1 and W_2) on section s ($s \in \bar{S}$). Formally,

$$x_{t,s} = \begin{cases} 1 & \omega_{t,s} \geq W_1 \\ 0 & otherwise \end{cases} \quad \bar{x}_{t,s} = \begin{cases} 1 & \omega_{t,s} \leq W_2 \\ 0 & otherwise \end{cases}$$

The train is affected by the disruption on section s, if and only if $x_{t,s} + \bar{x}_{t,s} = 2$.

The purpose of the model is to minimize the secondary delay of traffic running. The secondary delay of each train is the difference between the actual final arrival time (at the last section of the train), i.e., $\omega_{t,end}$, minus the sum of the planned entry time, the planned total travel time (without delay and disruption) and the primary delay of each train. Note that the secondary delay in our paper contains the delay caused by the propagation of the primary delays, and the impact of the speed limitation caused by the disruption, which refers to the new primary delay and its delay propagation. In other words, this is a comprehensive result of the interaction of primary delays, speed limits and railway capacity, and goes beyond the

Table 4
Speed level and additional running time.

Speed	250 $<v \leq$ 300 km/h	200 $<v \leq$ 250 km/h	160 $<v \leq$ 200 km/h	120 $<v \leq$ 160 km/h	$v \leq$ 120 km/h
Speed level ξ	$\xi=5$	$\xi=4$	$\xi=3$	$\xi=2$	$\xi=1$
Additional time span	$[\Delta t_5,\Delta t_4)$	$[\Delta t_4,\Delta t_3)$	$[\Delta t_3,\Delta t_2)$	$[\Delta t_2,\Delta t_1)$	$[\Delta t_1,M)$

traditional meaning of secondary delay. Formally,

$$\text{objective1} : \min : \sum_{t\in T} \omega_{t,end} - e_{t,s} - c_t - dt_t \tag{2}$$

We can remark that objective 1 includes slacks available in the timetable, and can use them to reduce the average total delay of trains. In other words, a train arriving early cancels the delay of a train arriving late by the same amount. As this is not the case in some other operational rules, we also consider an objective function where negative secondary delays are discarded. Formally, this reads:

$$\text{objective2} : \begin{cases} \min : \sum_{t\in T} y_t \\ y_t \geq \omega_{t,end} - e_{t,s} - c_t - dt_t t \in T, s \in R_t \\ y_t \geq 0 \quad t \in T \end{cases} \tag{3}$$

In the experiments, we will investigate the choice of the two objective functions towards the operations and the quality of the resulting schedule. In principle, objective 1 takes advantage of recovery time and buffer time in the timetable, to the extent that some trains might face some early arrivals. We remark here that trains are anyway not allowed to depart early from stations, thus the recovery times used are actually resulting in flexibility of arrivals, improving performance of timetable. In objective 2, only trains which have positive secondary delays are counted towards the performance measure: the performance of the operator is maximized. The optimizer can then only reduce (for what is possible) the delay of trains with positive secondary delay, exploiting flexibility and balance in the rest of the traffic to this extent.

We now express the main constraints of the problem in the following sub-sections.

4.2.1. Constraints related to running times *with different speed levels*

$$\omega_{t,q} - \omega_{t,p} \geq \delta_p \qquad \forall t \in T,\ p,q \in S,\ p,q \notin \cup\vartheta_\eta,\ p,q \in R_t,\ (p,q) \in F \tag{4}$$

$$\omega_{t,q} - \omega_{t,p} \geq \delta_p + \sum_{\xi=1}^{B_m} \Delta t_\xi \beta_{\xi,t,p} \qquad \forall t \in T, p,q \in \cup\vartheta_\eta, p,q \in R_t, (p,q) \in F \tag{5}$$

$$\omega_{t,q} - \omega_{t,p} < \delta_p + \sum_{\xi=2}^{B_m} \Delta t_{\xi-1} \beta_{\xi,t,p} + M\beta_{1,t,p} \qquad \forall t \in T, p,q \in \cup\vartheta_\eta, p,q \in R_t, (p,q) \in F \tag{6}$$

$$\omega_{t,q} - \omega_{t,p} \leq \delta_p + \Delta t_\xi \qquad \forall t \in T, p,q \in S, p,q \in R_t, (p,q) \in sw, \xi = 1 \tag{7}$$

Constraints (4) report the *minimum* operation (running) time for each section outside of open track areas. In other words, the running time is defined by the given permitted speed profile of a train running on the block section in the station area. Constraints (5) denote the minimum running time for those sections in the open track area, and a factor accounting for a variable speed level $\beta_{\xi,t,p}$ has to be included, where each speed level ξ corresponds an additional time Δt_ξ.The bigger the speed level ξ is, the faster the train speed is and the smaller the additional time $\Delta t_\xi$is. Constraints (6) constrain instead the *maximum* operation time (running time) on sections, to ensure a match between the speed level ξ and the actual running time required, within some thresholds. A typical range of extra running time related to each speed level is shown in Table 4, which can be calculated based on the empirical speed profiles of high-speed trains. Due to the combination of constraints (5) and (6), once the running time exceeds the threshold, the speed level will change automatically. Thus the final selected speed level ξ for each train on each section is determined, with the corresponding binary decision variable $\beta_{\xi,t,p}$= 1. Finally, constraints (7) report a maximum running time requirement for some sections where trains are not allowed to stop, for instance on switches. In this restriction, only the extra additional time Δt_1 is considered.

4.2.2. Entrance, stop and departure constraints

$$\omega_{t,s} \geq e_{t,s} + dt_t + \omega_0 \qquad \forall t \in T, (t,s) \in E_T \tag{8}$$

$$\omega_{t,q} - \omega_{t,p} \geq \delta_p + d_{t,p} \qquad \forall t \in T, q \in R_t, p \in D_t \tag{9}$$

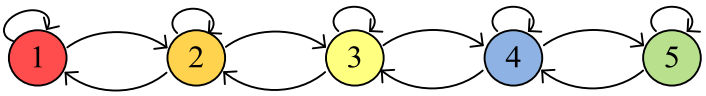


Fig. 4. Speed level transitions for two consecutive sections.

$$\omega_{t,s+1} \geq p_{t,s+1}+\omega_0 \qquad \forall t \in T, s \in D_t \tag{10}$$

We continue by reviewing the constraints related to *technical* activities, such as entrance, dwell and exit. All the entrance sections are linked to the dummy origin node by fixed arcs (the red solid arcs in the Layer 3 of Fig. 3) describing the expected entrance times of all trains. The start time of a dummy origin node ω_0 corresponds to a common origin of times, i.e., 0. Constraints (8) ensure that train t cannot start its journey before its planned entrance time $e_{t,s}$ plus the primary delay time dt_t and ω_0. Constraints (9) impose a minimum dwelling time $d_{t,p}$ on section (siding line) p at stations where a stop is planned, so that passengers have enough time to board or alight. The operation time on section $p$for train t should be not lower than the summation of real running time $\delta_p$and dwell time $d_{t,p}$. In addition to that, all the departure sections are directly connected to the dummy origin node by fixed arcs (the green solid arcs in the Layer 3 of Fig. 3) describing the planned departure time $p_{t,s+1}$, so constraints (10) enforce that each train cannot depart from a stop s (identified by operation $s+1$) before its planned departure time $p_{t,s+1}$. Trains without a planned stop are not considered in constraints (10).

4.2.3. Speed profile constraints

$$\sum_{\xi=1}^{\overline{B_m}} \beta_{\xi,t,s} = 1 \qquad \forall t \in T, s \in \vartheta_\eta, s \in R_t \tag{11}$$

$$\beta_{\xi,t,p} + \beta_{\psi,t,q} <= 1 \quad \text{if}(\psi > \quad \xi+1 \text{or} \psi < \xi - 1), \forall t \in T, p, q \in \vartheta_\eta, p, q \in R_t, (p,q) \in F, \xi, \psi \in B \tag{12}$$

$$\beta_{\xi,t,s} = 1 \qquad \forall t \in T, s \in \vartheta_\eta, s \in AD_t, \xi = 1 \tag{13}$$

This group of constraints expresses the *speed level* of trains related to the operations. The choice of the speed level ξ can be thought of as a restriction to the set of block sections in open track, for clarity. We recall here that the speed of train is classified into 5 levels and each level follows a minimum distance safety (braking distance or distance headway). We assume that, within some thresholds, the speed level in each section is constant. Different trains might have different maximum allowed speed levels $\overline{B_m}$, in other terms, different classes of high-speed trains might be given. The sum of $\beta_{\xi,t,p}(\xi = 1, 2, 3 \ldots \overline{B_m})$ in each section p is therefore equal to 1, as shown in Eq. (11). Constraints (12) are used to restrict the transitions of speed levels. The weight of the block section is fixed by the infrastructure design, and allows acceleration/deceleration only to the previous/next speed level. In other words, at most three combinations of speed levels are possible for one train to go across two consecutive sections: either keeping the same speed level ξ, or accelerating to the immediately higher level $\xi+1$, or decelerating to the immediately lower level $\xi-1$. We model the speed levels ξ and ψ over two consecutive block sections p, q as follows:$|\xi - \psi| \leq 1$. A graphic sketch of the speed level transition is plotted in Fig. 4, where the arcs denote the alternative transitions between speed levels, and number denotes the detailed speed level. Thus, for all combinations of speed level (ξ, ψ) from set $\{(\xi, \psi)||\xi - \psi| > =2, \xi, \psi \in B\}$, the values of $\beta_{\xi,t,p}$ and $\beta_{\psi,t,q}$ cannot be 1 at the same time, see constraints (12).

Finally, a train t might have extra constraints limiting its speed, typically in the vicinity of a station with a planned stop, or at the start or end of its departure or arrival route, if it is planned so. In this case, we use constraints (13) to fix the maximum speed level to 1, i.e., the minimum speed level, ensuring "normal" fixed block movement. In this way, the start speed level or end speed level after or before a stop for a train is reasonable and practical.

4.2.4. Speed limitation constraints

$$x_{t,s} \geq \frac{\omega_{t,s} - W_1}{M} \qquad \forall t \in T, s \in \overline{S} \cap R_t \tag{14}$$

$$\overline{x}_{t,s} \geq \frac{W_2 - \omega_{t,s}}{M} \qquad \forall t \in T, s \in \overline{S} \cap R_t \tag{15}$$

$$\sum_{\xi=1}^{B_m} \beta_{\xi,t,s} \geq x_{t,s} + \overline{x}_{t,s} - 1 \qquad \forall t \in T, s \in \overline{S} \cap R_t, \xi \in B \tag{16}$$

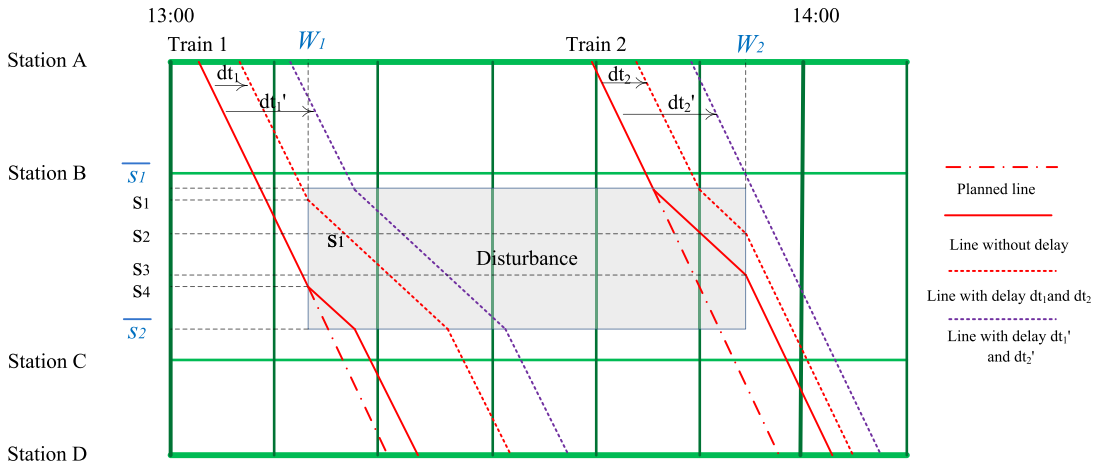


Fig. 5. Impact of a time-constrained disruption depends on the passing time of trains.

The above group of constraints relates to the running of a train in the disruption area: whether and to which extent, a train will suffer from the speed limitation. We use constraints (14) and (15) to determine dynamically whether a train will be affected at the time and space span $\bar{S}$ of the disruption. Specifically, due to the presence of primary and secondary delays, it is impossible to know on beforehand whether a train will pass into the disrupted area at the time of the disruption. We refer to Fig. 5, showing a typical time-distance diagram (time on the x-axis, increasing rightwards, space on the y-axis), with the disturbance reported as a shaded rectangular area in time and space; two trains (Train 1 and Train 2) are also involved. An increase of primary delay from dt_1 to dt_1' might greatly impact the run of Train 1, and make it run at a lower speed on a larger amount of impacted block sections. Conversely, an increase of delay dt_2 to dt_2' might change the passing time of train 2, and make it actually avoid the disruption. To identify those effects with relation to the (unknown) value of the precise passing time, we resort to the indicator variables $x_{t,s}$ and $\bar{x}_{t,s}$. The two variables determine whether the train runs after the start time W_1 of the disruption and before the end time $W_2$of the disruption or not, respectively, on the possible impacted sections. If $x_{t,s}=1$ and $\bar{x}_{t,s}=1$ at the same time, then the train is affected by the speed limitation on the section s considered. Furthermore, constraints (16) link the passing of each train during the disruption, with the possible limited speed levels B_m for the possibly impacted sections $\bar{S}$. Due to the uncertainty of entrance delay, all the trains which are scheduled to utilize in part or in full impacted sections are involved in the inequation. Thus, once the train is affected, i.e., $x_{t,s}+\bar{x}_{t,s}-1=1$, its speed level ξ will belong to the restricted set $\{\xi|\xi \le B_m\}$ and avoid any speed level in set $\{\xi|B_m < \xi \le \overline{B_m}\}$, otherwise, its speed level ξ can be anyone in the set B.

4.2.5. Capacity constraints

$$\omega_{j,q}-\omega_{i,p} \ge \delta_{p,q}-(1-\lambda_{i,j,p,q})M \qquad \forall i,j \in T,((p,q),(h,k)) \in A,p,k \in R_i,q,h \in R_j,i \ne j \tag{17}$$

$$\omega_{i,k}-\omega_{j,h} \ge \delta_{h,k}-\lambda_{i,j,p,q}M \qquad \forall i,j \in T,((p,q),(h,k)) \in A,p,k \in R_i,q,h \in R_j,i \ne j \tag{18}$$

We now describe the constraints to model the limited *capacity* of railway infrastructure at stations and on open tracks, regardless of the decision about speed; the normal headway rule applies in the fixed block signalling. As one section or any section(s) among conflicting sections cannot be occupied by two trains at the same time, alternative arcs are used to select a single order between the two trains. The (disjunctive) choice of the train orders is implemented by constraints (17) and (18) with a big-M structure. To be specific, if $\lambda_{i,j,p,q}$ is equal to 1, then constraint (17) becomes $\omega_{j,q}-\omega_{i,p} \ge \delta_{p,q}$ and constraint (18) is non-active with the expression $\omega_{i,k}-\omega_{j,h} \ge \delta_{h,k}-M$, which implies that the time when train j begins to use section q is at least $\delta_{p,q}$ time units later than the time when train i occupies section p, i.e., train i precedes train j. Otherwise, the constraint (18) is active and constraint (17) is non-active, enforcing the opposite order of trains.

4.2.6. Quasi-moving blocking constraints

$$\begin{aligned}&\omega_{j,q}-\omega_{i,p} \ge -(2-\mu_{i,j,\vartheta_\eta}-\beta_{\xi,j,q})M\\&\forall i,j \in T,\xi \in B,p,q \in \vartheta_\eta,q \in R_j,p \in R_i,i \ne j,p=q+\xi\end{aligned} \tag{19}$$

$$\begin{aligned}&\omega_{i,q}-\omega_{j,p} \ge -(1+\mu_{i,j,\vartheta_\eta}-\beta_{\xi,i,q})M\\&\forall i,j \in T,\xi \in B,p,q \in \vartheta_\eta,q \in R_j,p \in R_i,i \ne j,p=q+\xi\end{aligned} \tag{20}$$

$$M\mu_{i,j,\vartheta_\eta} \geq \omega_{j,f_\eta} - \omega_{i,f_\eta} \qquad \forall i, j \in T, f_\eta \in R_i, f_\eta \in R_j \tag{21}$$

The alternative arcs in Section 4.2.5 model the basic time-headway-based separation described by fixed block signalling. However, in a more general case of higher speed and headway based on quasi-moving block operations, we refer to constraints (19) ∼ (21), which also relate to the distance headway between trains and the capacity of the infrastructure. The distance headway to be kept free between two consecutive trains depends dynamically on the speed level of the following train. We thus require that at least ξ block sections should be kept free between two consecutive trains, when the second train runs at the speed level ξ ($\beta_{\xi,j,q} = 1$). Constraints (19) represent safety requirement related to distance headway, where train j runs behind train i and the speed level of train j is ξ. The start time $\omega_{j,q}$ of train j on section q cannot be earlier than $\omega_{i,p}$, where section p lies downstream of section q, being exactly the ξth section downstream. Similarly, the companion constraint, involving the opposite order, is expressed by constraints (20). Both constraints are again using the big-M structure to determine the order and the speed level chosen.

To avoid changing train orders on open tracks, we also include the final constraints (21). If the time ω_{i,f_η} when train i occupies the first section f_η of the open track ϑ_η is earlier than the time ω_{j,f_η} of train j on the same block section of the open track, then μ_{i,j,ϑ_η} is equal to 1, otherwise, it is 0. In a word, the train order decision variables are determined by the orders for the occupation on the first section of each open track, and the order of trains is fixed and constant in one entire open track area.

4.3. Two-step solution approach

In terms of complexity, the proposed MILP model has a huge number of constraints, growing quadratically with the amount of trains and the amount of block sections. Moreover, the amount of speed levels ξ considered also plays a role. This will put an increasing combinatorial load on the solution process, for which we devised an efficient and effective two-step solution process. This is used to determine quickly an initial solution by fixing some variables related to the speed profile, and strengthen the bounds for improving the computation speed and quality of the solution of the general problem.

More precisely, the procedure works by fixing the speed level of each train to the maximum speed possible, in either disrupted or undisrupted sections. This means that trains are accelerating as fast as possible after a planned stop, and decelerating as fast as possible before any planned stop. This is modelled by Eq. (22)–(24), where the first two equations are used to fix the maximum speed for trains which are not subject to the disruption in any case on sections which are not subject to the disruption. The third equation is used to fix the maximum speed limitation for those trains on sections which can be possibly impacted by the disruption. We simplify the disruption by extending the disruption area ($\widetilde{S}$) to allow acceleration and deceleration, and we also fix a-priori the amount of trains (T_0) which are likely to be affected by the disruption, based on the original timetable. In this sense, we thus neglect the actual primary and secondary delay to determine whether a train is impacted by the disruption or not. When the additional constraints (22)–(24) are added to the original model in Section 4.2, a solution can be calculated, which is always a feasible solution for the original model. Considering this upper bound solution allows to speed up considerably the search process, as we will report in the next section.

$$\beta_{\xi,t,s} = 1 \qquad \forall t \in T - T_0, s \in \cup G_\vartheta, \xi = \overline{B_m}, s \in R_t \tag{22}$$

$$\beta_{\xi,t,s} = 1 \qquad \forall t \in T_0, s \in \cup G_\vartheta - \widetilde{S}, \xi = \overline{B_m}, s \in R_t \tag{23}$$

$$\beta_{\xi,t,s} = 1 \qquad \forall t \in T_0, s \in \widetilde{S}, \xi = B_m, s \in R_t \tag{24}$$

5. Computational experiments

We solve the proposed model by the commercial solver IBM ILOG CPLEX 12.6.0, with default parameters. Our experiments are all executed on a machine with an Intel(R) Core (TM) i5-4690 processor CPU 3.50 GHz and 8.00GB memory. We divide our experiments into two parts, first a more extensive study on a real network with (realistic but fictitious) mixed-speed traffic, and then an application on a larger real network with the timetable used in operations, which schedules trains with a single maximum speed.

As Chinese high-speed railway lines are all double-track lines, the two tracks in different directions are completely independent for what concerns operations. We therefore limit our study to a single direction within a corridor-like network. Due to the layout of the lines, we also do not consider rerouting actions. We also consider all trains have the same dynamic characteristics, i.e., the same acceleration/deceleration curve, depending on the speed.

We compare our proposed method with three benchmark solutions, including the First-Scheduled-First-Served solution (FSFS), the First-Come-First-Served solution (FCFS) and the Time-Headway-Only (THO) solution. The first corresponds to deciding the orders of the trains based on the original schedule, i.e., keeping the order planned in the original timetable. The second decides dynamically the order of trains based on their arrival time at a station where an order has to be decided; the train which arrives first, will pass first. Apart from those two scheduling rules, we also report the solution considering a *fixed* time headway between two trains, with the aim to demonstrate the advantage of variable time/space headway

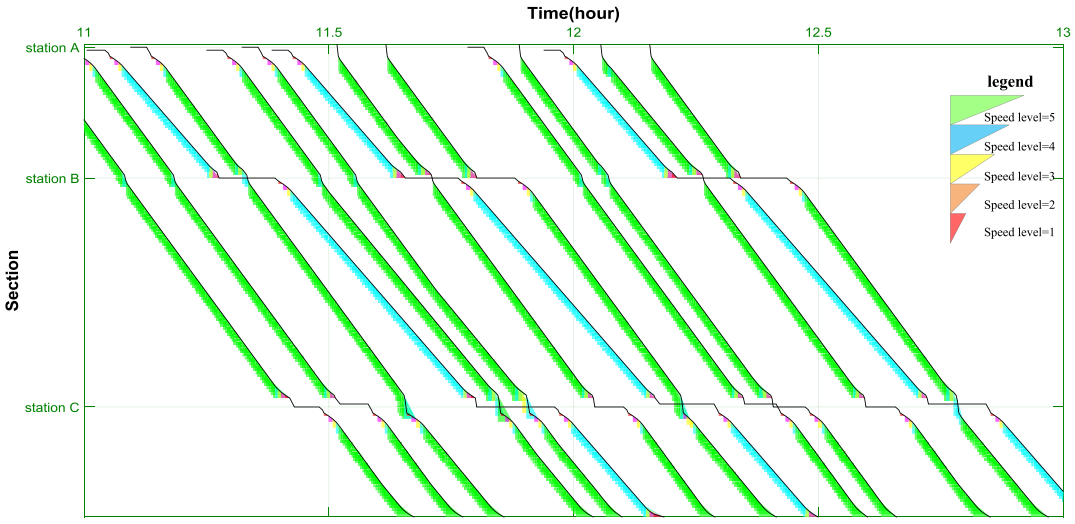


Fig. 6. Planned (original) timetable.

Table 5
Indices of disruption cases.

	$c=\|S\|=20$	$c=\|S\|=30$	$c=\|S\|=40$	$c=\|S\|=50$	$c=\|S\|=60$
$b=B_m=1$	1	2	3	4	5
$b=B_m=2$	6	7	8	9	10
$b=B_m=3$	11	12	13	14	15
$b=B_m=4$	16	17	18	19	20

when dealing with high speed traffic and quasi-moving block signalling system. We finally consider the solution which is computed by relaxing (/neglecting) the constraints corresponding to limited railway capacity, as a lower bound which can be useful to evaluate the solution quality of the models, as well as the delay time caused by conflicts.

5.1. Realistic Chinese high-speed network with heterogeneous traffic

5.1.1. Test case

In this section, we test the proposed model on a timetable considering heterogeneous high-speed traffic, where trains have different stopping patterns and also different maximum speeds. More precisely, a time horizon of one and a half hour, planning 11 higher speed trains (300 km/h) and 3 slower speed trains (250 km/h), is considered. The infrastructure is a rail line with 3 stations and a total of 160 sections, whose length is 1360 m per section, commonly used in a typical high speed design. The planned timetable is plotted in Fig. 6, where the colour of the line indicates the speed allowed, similarly to Fig. 3. Thus, the green lines correspond to the runs of the faster high-speed trains (300 km/h) and the blue lines correspond to slower high-speed trains (250 km/h). It is also possible to see how the speed decreases or increases before and after a planned stop.

We consider 8 delay cases, namely the timetable without any delay (the first case), and 7 more cases with primary delay. The primary delays affect all trains running, and follow a 3-parameter Weibull distribution, as in Corman et al. (2011b). The probability distribution of delays for the higher speed trains has the following parameters: $\beta_h=$ 1.48, $\eta_h=$ 560, $\gamma_h=$ 205, while for the lower speed trains is $\beta_s=2.27$, $\eta_s=394$ and $\gamma_s=315$. The average total primary delay (among the 7 cases) is about 3381 s. We moreover consider a set of 40 disruption cases, with varying time spans of the disruption, degrees of speed limitation (i.e., maximum speed allowed), and amount of sections affected by the disruption. In what follows, we report the results on the 40 disruption cases, in the case of delayed / punctual traffic and according to the two objective functions proposed. The results concerning primary delay are always the average value of the 7 delay cases. When necessary, we identify the disruption cases by the triple $\langle a,b,c\rangle$ where a represents the duration, in hours; b represents the degree, as maximum speed level allowed B_m (the higher, the lighter the disruption, the lower, the stronger the disruption), and c reports the amount of sections affected |S|. We identify a triple $\langle *, b, c\rangle$ with disruption indices, which are reported in Table 5, and will be used to report schematically the disruptions in the subsequent figures.

5.1.2. Comprehensive influence of speed limitation and primary delay

Fig. 7 shows the relation between the number of sections impacted by the disruption (x-axis), severity levels of the disruption (different colours in curves), primary delays (solid/dashed), and secondary delays (y-axis) of objective 1. The plot

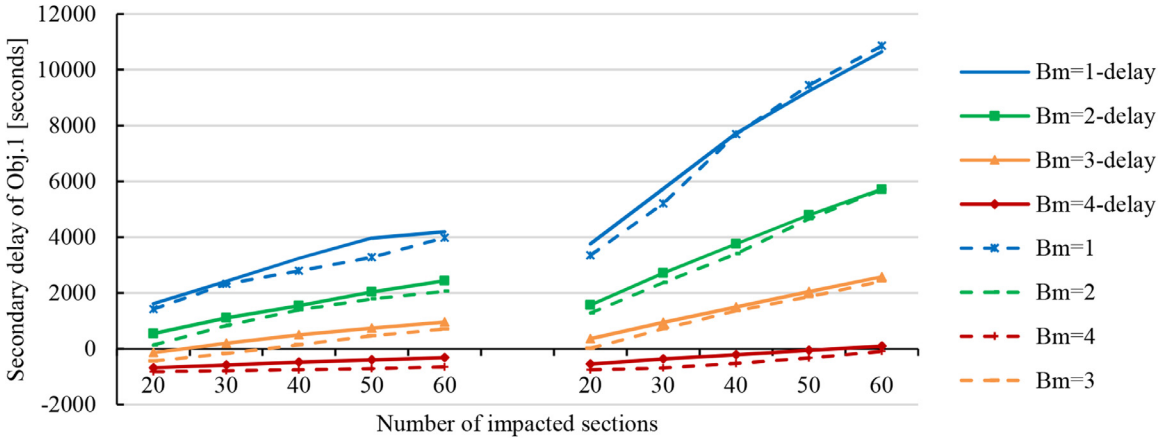


Fig. 7. Secondary delay caused by different speed limitations, primary delay, impacted sections (left = 0.5 h; right = 1 h). (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

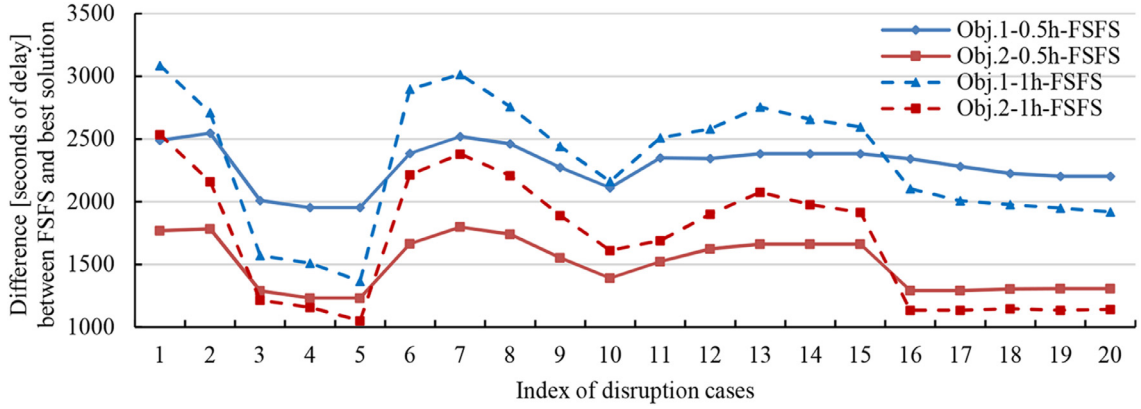


Fig. 8. Analysis of the difference between the FSFS solution and the best solution, with primary delay.

at the left-hand of the figure concerns a disruption of half an hour, while the right-hand part reports the results with 1 h disruption. Dashed lines indicate the cases without primary delay, while solid lines report the cases with primary delays. The objective function considered and reported is objective 1 (i.e., negative secondary delays are allowed).

From the analysis of Fig. 7, it is evident how the secondary delay increases with the decrease of speed limitation and the increasing amount of impacted sections. There is a linear trend between the numbers of impacted sections and secondary delays under the disruptions with same duration. However, there are different increasing rates related to different durations, see the left-hand part and right-hand part in Fig. 7. In detail, the slope of the blue line ($B_m = 1$) on the right-hand plot in terms of [delay seconds/impacted section] is about 3.5 times higher than that of the left-hand plot, while their durations of the disruption only double. The gap between lines with different speed limitations grows with the increasing disrupted sections. Comparing each dashed line (case without primary delay) with its corresponding solid line (cases with primary delay), the impact of primary delays on top of the disruption can be assessed. A very severe speed limitation weakens the (relative) influence caused by primary delay, as can be understood by the fact that the (right) ends of the dotted/solid lines in Fig. 7 touch each other.

This figure analyses the most important parameters of speed limitations, and the relationship with secondary delay. The linear relation between the amount of disrupted sections and the total amount of delay is rather easy to be measured, but the level of speed limitation and the duration have more complicated effects. In fact, the recovery time in the timetable can even absorb delays caused by a minor speed limitation, e.g., for $B_m = 3$ or 4, some negative secondary delays are found by objective 1, if the duration and the amount of disrupted section are relatively small. In fact, the timetable proves to be robust enough to deal with a speed limitation level 4, and no extra delays are generated, if objective 1 is considered.

5.1.3. Benefits of optimization

We now benchmark the results of the proposed model against three different scheduling rules. Firstly, the First-Scheduled-First-Served (FSFS) scheduling rule keeps the order as planned in the timetable, and adjusts the speed based on the speed limitation and interaction between trains. We consider 7 cases with primary delays only. Fig. 8 reports graphi-

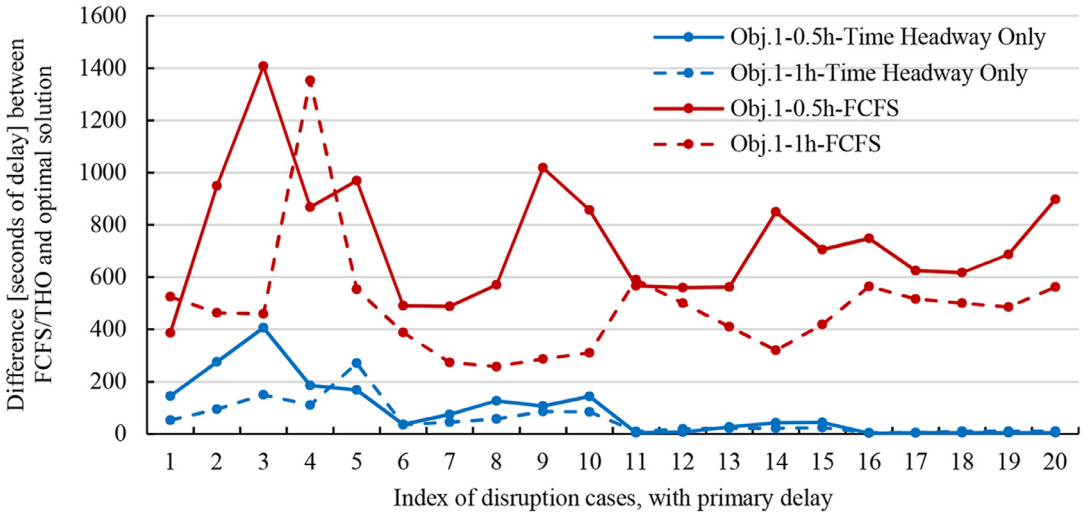


Fig. 9. Analysis of the difference between FCFS /THO solutions and the best solution, with primary delay.

cally the difference (objective 1, in blue, objective 2, in red) between the objective value of our proposed optimization model and the FSFS value. The cases are divided into half an hour disruption (solid lines) and one hour disruption (dashed lines). The x-axis shows the 20 cases of disruptions, as indexed in Table 5. The optimized solutions achieve at least a 1300 and 1000 s reductions, with the optimization of objective 1 and objective 2 respectively; this corresponds to 10%–50% of the total delay. Note that we are always able to improve at least 1000 s total delay with regard to the FSFS benchmark. The (almost constant) gap (500∼700 s) between the lines corresponding to objective 1 and objective 2 depends on the amount of recovery time that can be exploited in the operations. The solid (blue/red) lines are below the dashed (blue/red) lines mostly, except for cases 3–5, and 16–20. This refers to two different effects. The first is that a strong speed limitation (disruption case 3, 4, 5 with one hour duration) leads to a smaller improvement by adjusting train orders, due to the sharp decrease of railway capacity. In other words, when the speed limitation is very strong, many of the delays come from the speed limitation, and cannot be easily avoided by rescheduling. Instead, when a light disruption (case 16–20) is present, a longer disruption gives more opportunity to improve the schedule by optimization, and the FSFS performs worse.

We also test two other benchmark solutions, in Fig. 9. One is the First-Come-First-Served (FCFS) scheduling rule on the same disruption cases. This rule myopically looks at the entrance time of each train in a station, while neglects the future and possible conflicts related to different maximum speeds. The other one is labelled *Time-Headway-Only* (THO in short), and is reported with the aim to illustrate the strength of variable distance headway. In the THO model, a constant time headway between consecutive trains is considered, which is not dependent on the speed. This would be the easiest extension of fixed block models to high-speed traffic. We consider this headway to be 180 s, which is widely used in practice. The solution computed by using this fixed time headway is then recast into feasible high-speed movements, which follow the speed profiles and regulations in the other solutions. The differences between the best solution of the proposed model, and the FCFS or THO solutions are reported in Fig. 9. Red lines relate to FCFS, blue lines refer to THO. Solid lines refer to half an hour disruption, and dashed lines refer to one hour disruption. We remark here that due to very similar values between objective 1 and objective 2, and for the sake of clarity, we do not plot the relative results of objective 2 in Fig. 9. All the positive values reported in Fig. 9 show the advantage of our model compared to the benchmarks. In fact, compared to the FCFS solution, the optimization reduces delay by 20% to 40%, and always by at least 600 s. This makes it clear that a dynamic scheduling rule like FCFS improves over the static timetable order, but is still well below the potential of optimized operations. Compared to the model reported as Time-Headway-Only, the gaps are substantially smaller. In fact, the THO model also optimizes, but fails in representing correctly the dynamics of interaction between successive trains, and the triggering of deceleration when not enough space is available. Such a model results in 4% to 12% more delays, compared to the precise modelling of distance headway of the proposed model. The maximum difference reaches up to 406 s, for the most serious disruptions; for the light disruptions, where scheduling is less crucial, this is smaller, around 30 s. Considering together optimization of train traffic and speed level enables to achieve the best solution in all cases, outperforming all three benchmarks considered.

5.1.4. Factors of performance

We here systematically identify the factors which evaluate the performance of the optimization model. We start by introducing a reference lower bound for each factor, and computing the gap with this lower bound. The first study focuses on *conflicts*; the lower bound thus neglects all delays caused by conflicts. More precisely, the lower bound is computed as the optimal solution of the model obtained by relaxing all capacity-related constraints, i.e., trains can overtake each other anywhere, when needed (we refer to this lower bound as a *free flow model*).

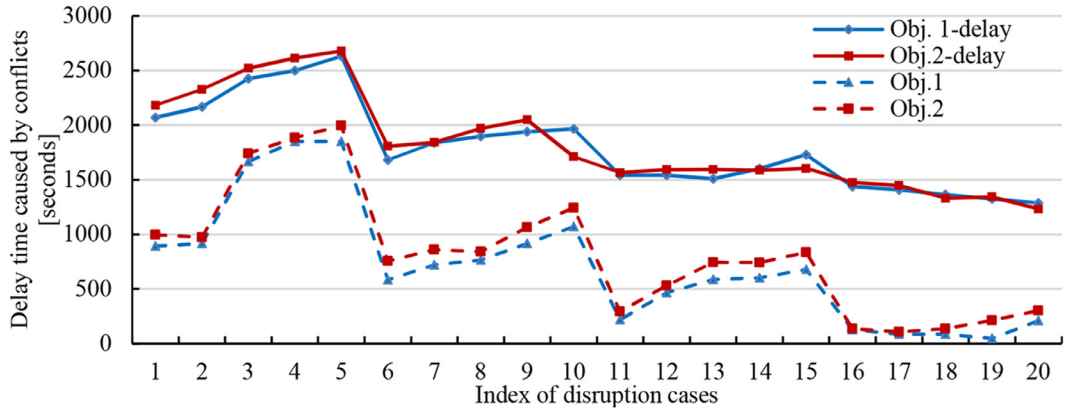


Fig. 10. Delays caused by conflicts, with 1 h speed limitation.

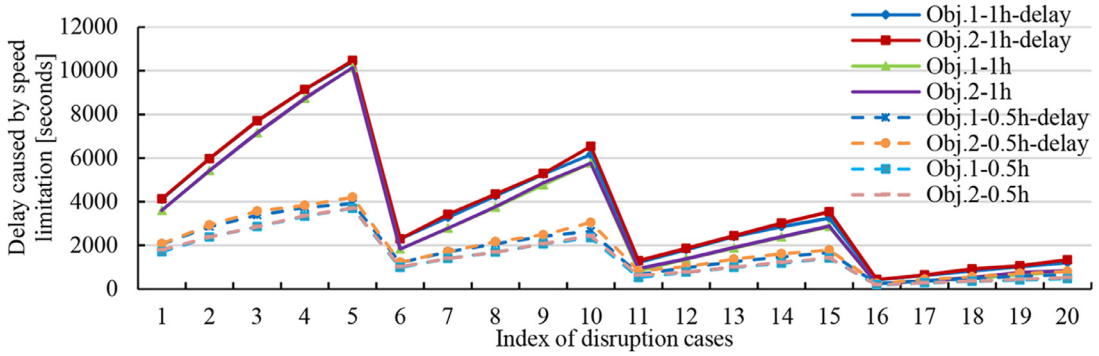


Fig. 11. Delays caused by speed limitations. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

The difference between this lower bound and the best solution clarifies the extent caused by conflicts. We report both differences of objective 1 and objective 2 (the delay caused by conflicts) in Fig. 10, limited to the cases with 1 h disruptions. The solid lines are delays due to conflicts related to cases with primary delay; the dashed lines refer to cases without primary delay. The y-axis shows the difference (in seconds) between the objective values of the proposed optimization model and the lower bounds, for each disruption case. The positive value associated with all disruptions means that (obviously) the optimization model performs worse than the lower bound. In fact, what is reported in Fig. 10 is the delay times caused by conflicts. The gap between the averages of the solid lines and dashed lines, which is about 1000 s, is the effect of the delay propagation due to the primary delays. There is about one third of the initial primary delays (which are 3381 s on average). Another trend evident from the figure is a decreasing trend with an increasing disruption index; the maximum is (as expected) found in case 5, with the strongest speed limitation, and the widest geographical extent. This relates to the fact that a stronger disruption (a lower disruption index) perturbs more traffic, more heavily, thus deviating traffic further away from the plan. In those cases, the amount of delay propagation can reach about 2800 s for objective 1 and objective 2, see the highest peak in Fig. 10. In fact, both objectives deliver basically the same results, which demonstrate that both approaches deliver similar solutions in terms of structure.

We next move on studying the impact of the *speed limitation*. In this case, we compute a different reference lower bound, where both the capacity constraints (i.e., constraints (17) and (18)) and speed limitation constraints (i.e., constraints (14)–(16)) are relaxed. The final result of this reference lower bound is −2404 s with primary delay and −1135 s without primary delay. The influence of speed limitation can be measured by the gap between this bound and the result of the free flow model, which is displayed in Fig. 11. Again, the solid lines correspond to the 1 h speed limitation and the dashed lines correspond to the 0.5 h speed limitation, with different objectives and primary delays.

Fig. 11 shows how the delay caused by the speed limitation increases linearly with the extent of the disruption and with the severity of the speed limitation. The curves with and without primary delays are close to each other, separated by a small gap (300 s). This means that the primary delay play a minor role in the effect of speed limitation, by changing and enlarging the set of trains impacted. The solid lines show a sharper increase than that of the dashed lines, which demonstrates the stronger effect of a longer disruption duration. In fact, the slope of the solid line is about 3 times of that of the dashed line, which is larger than the increase (double) in terms of disruption duration. Recall that for the model considering capacity constraints and speed limitation, the slope is about 3.5; the difference gives an idea of the extent by

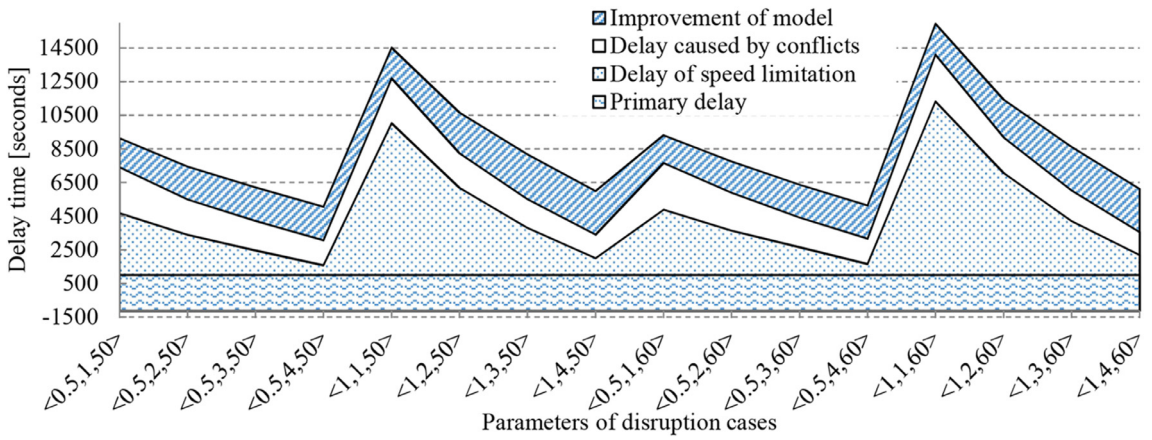


Fig. 12. Delay component of objective 1 under speed limitation.

which the speed limitation results in additional delays. The maximum delay can reach up to 10,000 s, for disruption index 5; for the lightest disruption cases, i.e. 16–20, the delays are almost negligible. The two objectives result in basically the same values, i.e., the delay caused by speed limitation can be considered as an unavoidable disturbance, which cannot be avoided by the optimization algorithms.

We summarize the factors investigated in Fig. 12, which traces the delays observed to their sources. We consider to this extent the reference values we computed so far, focusing on speed limitation, the improvement caused by optimization (FSFS vs. the proposed model), primary delays and conflicts. While some of those sources are constant in their effects, some others are strongly dependent on the disruptions and on their effects. A selection of 16 disruption cases is reported. The layer at the bottom of the Fig. 12 (with wave pattern) denotes the influence of primary delay: due to the available slack (about 1500 s), the 3866 s of primary delays in total result in about 1000 s final delay. The upper layer just above it (dotted) shows the delay time related to speed limitation, which occupies a large proportion among the total delay. This delay time grows proportionally with the duration, severity of speed limitation, and number of impacted sections. The layer further on top (white) depicts the delay propagation due to unavoidable conflicts caused by primary delay and speed limitation. This can be quantified in roughly extra 2000 s. Finally, the top layer (with shaded pattern) illustrates the benefits of optimization measures (retiming & reordering), which are around 2000 s. To summarize, our approach manages to reject about half of the consecutive delay caused by conflicts under the combined influence of speed limitation and primary delay.

5.1.5. Detailed analysis of a solution: scheduling and speed management

We now sketch graphically some details of the solutions found for the disrupted case $\langle a, b, c\rangle = \langle 0.5\,\text{h}, 1, 50\rangle$ with primary delay (3866 s). The solutions computed by the proposed model and FSFS model using objective 1 are reported respectively in Fig. 13 and in Fig. 14. The most interesting parts for readers are identified by the area in purple circles in the figures. Conflicts are identified by lines which have colour less than the maximum planned speed, respectively green and blue. Those are mostly related (indirectly) to the speed limitation, or situations where trains come too close to each other. The optimized solution avoids a part of such conflicts by adjusting trains speed to prolong or reduce the running times of train on beforehand, changing the order of trains already well before they enter the impacted area, making good use of railway infrastructure capacity. In general, the optimization approach is able to bundle trains with slower speed after trains with higher speed, to improve the throughput and reduce the overall delay. In this way, slower speed trains face the speed limitation later or the end of the disruption. In fact, keeping a slower speed train along its planned order might result in delays for many faster trains after it, when trains will get out the speed limitation area (see the circle at the right bottom of Fig. 14). Above all, we identify two main effects, namely an overtaking between trains of the same category, which allows trains to keep a better performance, and an overtaking between trains of different categories, which mitigates the influence of speed reductions and conflicts.

5.1.6 Train control in terms of speed profile

One innovative possibility of the proposed model is to integrate the management of speed while optimizing traffic. In this section, some insights on how this works in practice is shown in Fig. 15, which is a zoom of the disruption case $\langle 0.5\,\text{h}, 1, 50\rangle$ without primary delay, where two successive trains are coordinating their speed. We show the situation when both trains are exiting the disrupted area and are thus theoretically allowed to accelerate to full speed. In fact, this is hindered by a very small distance headway between the two trains, and moreover their maximum speeds are different. In Fig. 15, the blue train precedes the other train, which has mainly a green colour. For this latter train, there is some oscillating behaviour (blue/green, in one case reaching yellow), when the distance headway between them goes below the amount of free sections related to the speed followed. In general, it might be interesting to have some extra optimization model

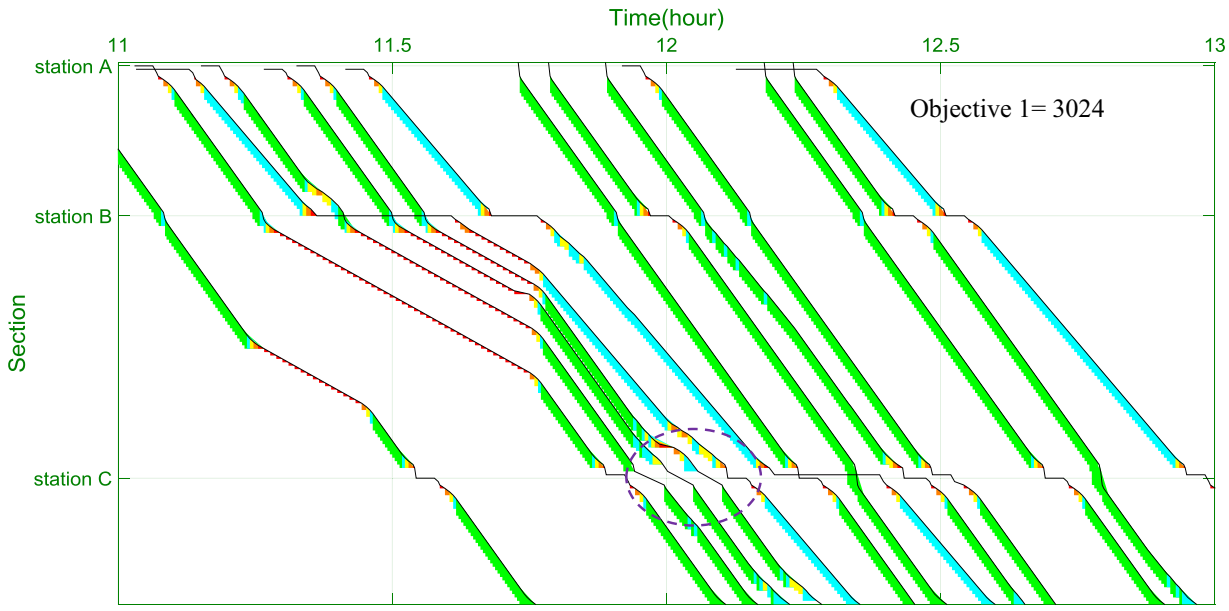


Fig. 13. Final rescheduling solution of case ⟨0.5 h, 1, 50⟩ and delays, optimized with objective 1.

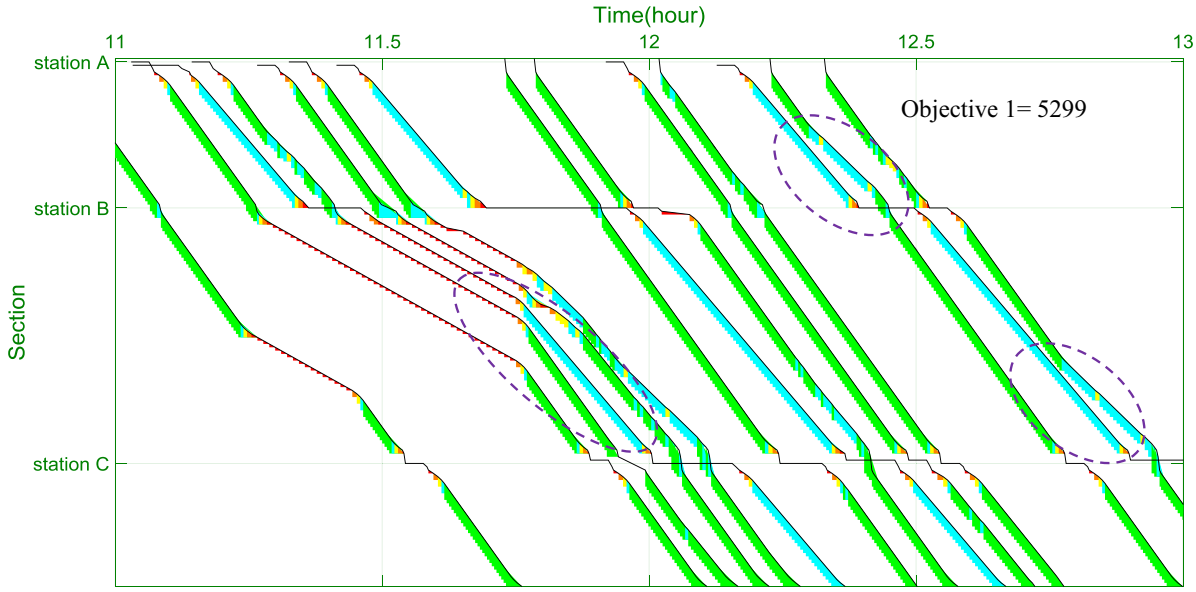


Fig. 14. Final rescheduling solution of case ⟨0.5 h, 1, 50⟩ and delays, with FSFS.

which aims at reducing the nervousness of control actions (see Corman and Quaglietta, 2015), when put in closed loop, for instance, reducing delays at once, and going with a more constant speed level.

5.2. Real Chinese high-speed railway with homogeneous traffic

5.2.1. Test case

A different test case is performed in this part, which corresponds to a real high-speed network in China connecting Wuhan-Guangzhou. It focuses on a railway traffic control post from E-Hengyang (included) to the North of Guangzhou (excluded). Figs. 16 and 17 show a scheme of the track layout and a part of the timetable. The former presents 6 railway stations considered, and 6 open track areas between them, spanning about 500 km. The total number of block sections involved is 374, and the length of each section in open track is 1360 m. The red shaded rectangles are the locations of the two speed limitations considered, which have further different extents in time, and severity, as determined later.

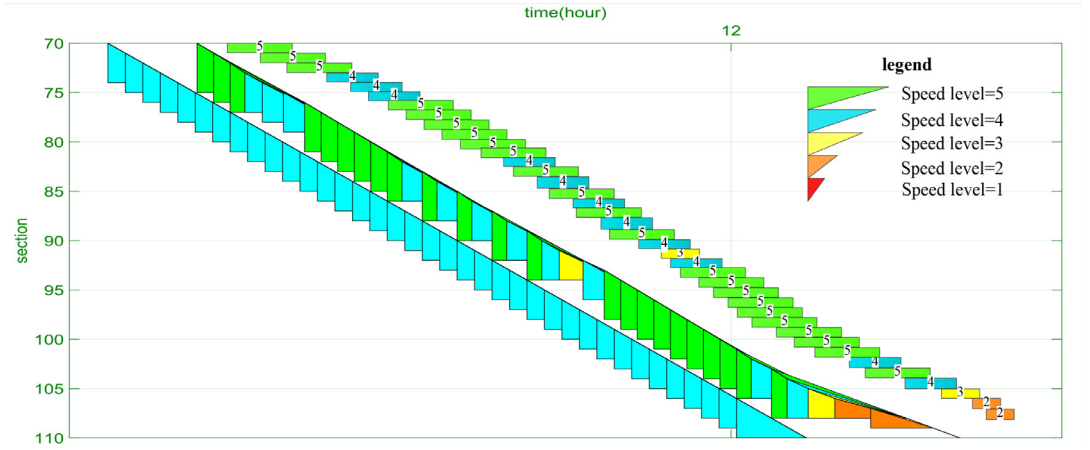


Fig. 15. Conflict-related minimum distance headway. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

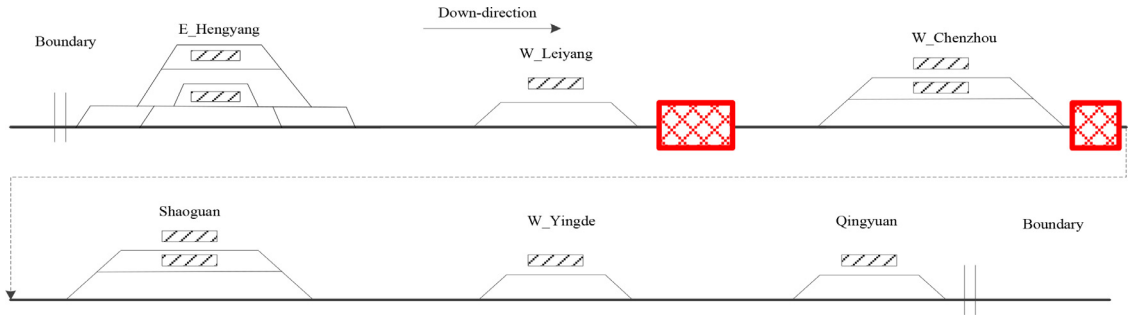


Fig. 16. Layout of the railway line Hengyang-Guangzhou.

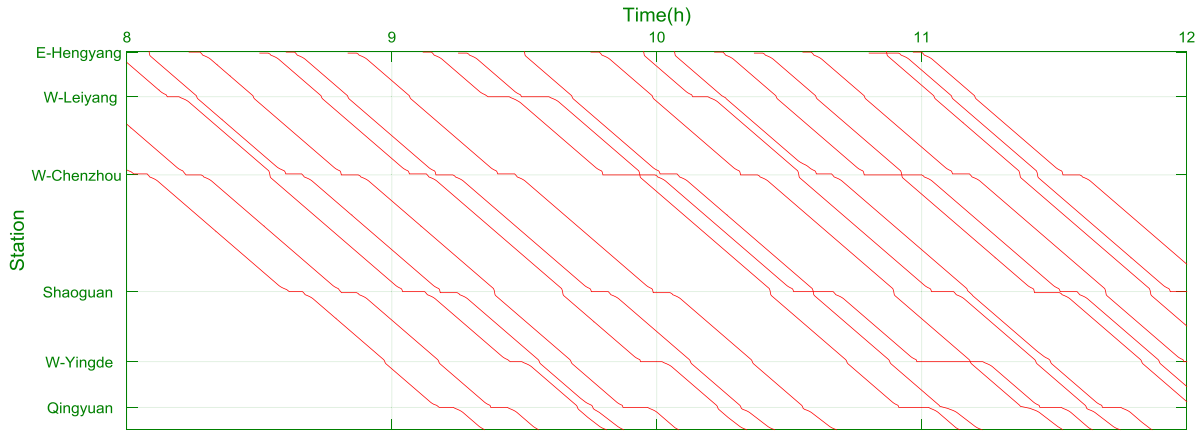


Fig. 17. Planned timetable of the railway line considered.

The timetable used is the one currently in operation, which schedules 98 high-speed trains per day on this part of the network, and all high-speed trains are with the same maximum speed (300 km/h). We restrict the investigation to the trains running on the time horizon between 9:00 and 11:00, which considers 20 trains displayed in Fig. 17. The running time depends on the speed level, as explained in Table 6: at full speed, traversing a typical block section takes 16 s, while for slower speeds the additional running times Δt_{ξ} are reported in Table 6. We consider 20 cases of primary delays, affecting all trains, based on the same Weibull parameters for high-speed traffic considered in previous test cases, namely (β_h= 1.48, η_h= 560, γ_h= 205). We report on the average over those 20 cases, when not stated otherwise.

The disruptions considered are the combination of a varying duration (0.5 h, 1 h and 1.5 h time span, all starting from 9:00), four kinds of speed limitation levels ($B_m = 1, 2, 3, 4$), and disruption extent (50 sections disrupted between W-

Table 6
Additional running time for each speed level.

Basic running time	Speed level	5: 300–250 km/h,	4: 250–200 km/h,	3: 200–160 km/h,	2: 160–120 km/h	1: 120 km/h or less
16 s	Additional running time (seconds)	[0,3)	[3,8)	[8,14)	[14,24)	[24,+ ∞)

Table 7
Disruption indices.

	$c=\|S\|=50$ a = 0.5 h	$c=\|S\|=50$ a = 1 h	$c=\|S\|=50$ a = 1.5 h	$c=\|S\|=60$ a = 0.5 h	$c=\|S\|=60$ a = 1 h	$c=\|S\|=60$ a = 1.5 h
$b=B_m=1$	1	5	9	13	17	21
$b=B_m=2$	2	6	10	14	18	22
$b=B_m=3$	3	7	11	15	19	23
$b=B_m=4$	4	8	12	16	20	24

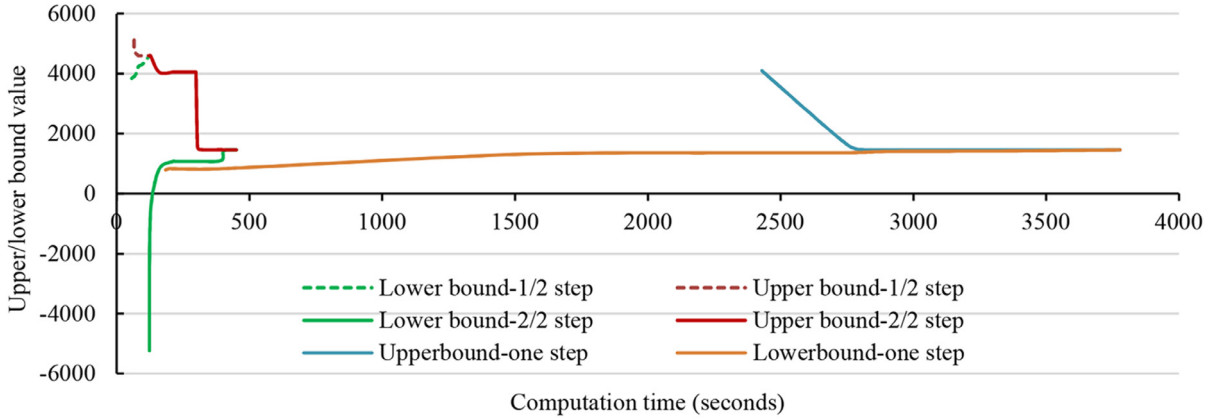


Fig. 18. Comparison of computation times with one-step method and two-step method. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

Leiyang → W-Chenzhou or 60 sections disrupted between W-Chenzhou → Shaoguan), which can be summarized by a triple ⟨*a, b, c*⟩. This determines a set of disruption indices, which are reported in Table 7, and will be used to identify schematically the disruptions in the following figures.

5.2.2. Solution approach

The number of variables and constraints depends quadratically on the amount of block sections and trains, and the extent of speed limitation, while it does not depend on the duration of the disruption and its severity. There are 20 trains, running over 374 sections, leading to 48,519 variables and 502,208 constraints. The large amount of variables and constraints determine strongly the computational complexity, which shows the essential value of the two-step solution method to speed up the solution process.

Fig. 18 illustrates the performance of computing upper and lower bounds of the problem along the computation time, when solving the problem with a disruption ⟨ 1.5 h, 1, 50⟩, from 9:00 to 10:30, 50 impacted sections, and objective 1. The red lines refer to the upper bounds of the two-step solution approach, namely the first step is red dashed, and the second step is red solid. The lower bounds for the two-step approach are in green, also here dashed green is the first step, and solid green is the second step. The first step considers a fixed speed to find an initial solution (with value 4300 s), which is achieved (and proven optimum) within 120 s. This solution is used as starting upper bound for the second step, which then converges later to a value around 1500 in about 300 s. The solid orange/blue lines depict respectively the lower bound and the upper bound of a single step solution approach (without the initial solution or upper bound). This approach requires about 2500 s to get the first feasible solution, which is then improved just before 3000 s. The lower bound improves slowly, with the optimality gap closed by one hour of computation time. In comparison, the two-step approach requires one eighth of the time to find the same optimal solution. Based on other experiments, we find out that CPLEX is in fact being able to find the optimal solution reasonably quickly, but has troubles in proving its optimality (even starting from the optimal upper bound, more than half an hour is required to close the search). We therefore restrict the maximum computation time to 600 s, and report the solution computed by that time, which is still acceptable for real-time usage. The first step of computation is limited by a maximum computation time of 150 s.

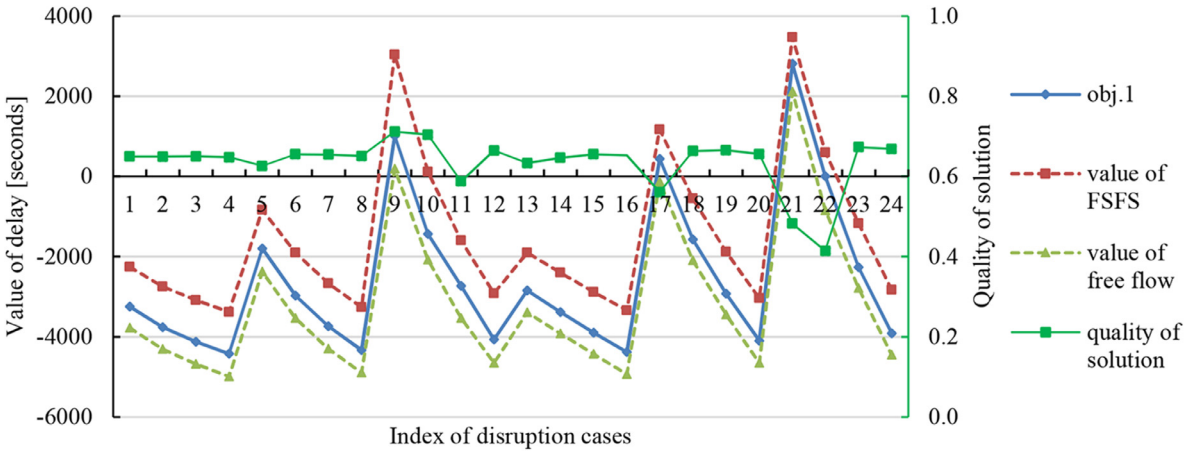


Fig. 19. Relative results analysis of objective 1.

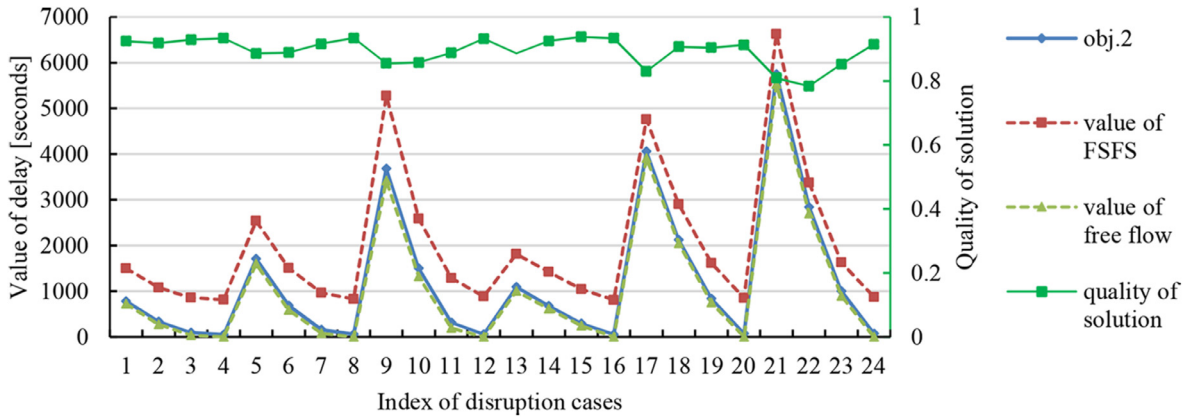


Fig. 20. Relative results analysis of objective 2.

5.2.3. Impact of speed limitation

We now evaluate the impact of speed limitation (24 disruption scenarios) towards performance of traffic, according to the two objectives, a benchmark solution (FSFS), and a lower bound solution (free flow model). The relationship among the objective value (blue solid line), value of FSFS (dotted red line) and value of free flow model (dotted green line), and moreover a synthetic indication of quality of the solution, based on the gaps (solid green, referring to the right-hand axis) are reported in Figs. 19 and 20, respectively for objective 1 and objective 2. The relative values of each case come from the average over the 20 primary delays. The index along the x-axis denotes the disruption cases in Table 7. The secondary delay has a non–linear relation with the duration, based on the trend of the blue lines in Figs. 19 and 20. There is a linear increase in delays with the decrease of speed limitation when the duration is 0.5 h for both objectives, even though in general there is an exponentially growing trend for longer disruptions, at speed level 1.

The final value of the optimization model (blue line) is always better than the FSFS benchmark, and is very close to the performance of the free flow model which neglects capacity. In any case, this limited gap is dependent only to a minor extent on the duration and extent of the disruption, as well as speed limitation: the optimization approach manages to deliver good solutions consistently. In fact, the optimization approach, by changing train orders, can reduce the delay successfully even in the case of strong speed limitation. To make this more evident, Figs. 19 and 20 report a quality index, in a solid green line. This is an index describing the quality of the solution, and practically computed as *quality coefficient* = (*value of FSFS* − *final value*)/(*value of FSFS* − *value free flow*). The larger the coefficient is, the better the quality of the final solution is, with regards to both the lower bound and the benchmark. This metric reports how much of the gap between the FSFS and the lower bound can be reduced by optimization; a value of 1 means reaching the lower bound (which is not tight), and a value of 0 means performing as bad as the timetable order. For objective 1, the coefficient fluctuates between 0.6 and 0.7, with the exception of the strong disruption 22, where a minimum quality of 0.4 is reached. In other terms, the gap can be generally reduced by 70%. For objective 2, a consistent level of 0.9 is achieved for the quality index; in other words, the gap can be reduced by as much as 90%.

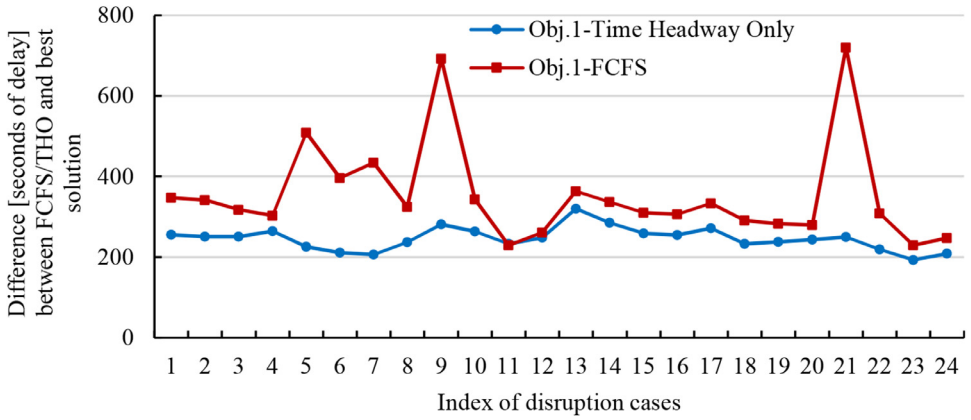


Fig. 21. Analysis of difference between FCFS/THO solution and the best solution.

Table 8
Computation time and optimality gap.

Index of case	Objective 1				Objective 2			
	Computation time (time limit = 600 s)	Amount of time limits reached (at 600 s)	Average optimality gap/average best solution/average optimal solution (a/b/c)*	Gap between optimal solution and best solution with 180 s time limits	Computation time (time limit = 600 s)	Amount of time limits reached (at 600 s)	Average optimality gap/average best solution/average optimal solution (a/b/c)*	Gap between optimal solution and best solution with 180 s time limits
1	172.52	0	0	0.35%	119.69	0	0	0%
2	133.01	0	0	0.01%	101.71	0	0	0%
3	115.13	0	0	0%	79.89	0	0	0%
4	101.15	0	0	0%	88.18	0	0	0%
5	492.37	8	8.18%/−1864/−1902	15.22%	468.84	2	9.97%/1654/1642	27.21%
6	264.31	0	0	0.94%	175.89	0	0	2.27%
7	211.72	0	0	0.04%	129.75	0	0	0.13%
8	122.78	0	0	0%	101.20	0	0	0%
9	577.74	13	40.38%/1041/997	203.47%	596.4	9	6.71%/3745/3642	20.82%
10	356.13	1	41.4%/−1368/−1545	18.04%	317.88	0	0	7.99%
11	276.04	0	0	1.14%	161.89	0	0	2.05%
12	129.30	0	0	0%	101.21	0	0	0%
13	200.36	3	11.63%/−2492/−2507	3.39%	362.44	0	0	18.86%
14	224.91	0	0	0.70%	220.45	0	0	5.56%
15	109.46	0	0	0%	81.98	0	0	0.10%
16	76.82	0	0	0%	66.35	0	0	0%
17	253.05	1	23.87%/1312/1232	19.87%	370.39	0	0	0.73%
18	235.01	0	0	0.17%	198.76	1	1.47%/1718/1718	0.33%
19	169.18	0	0	0.03%	72.01	0	0	0%
20	103.92	0	0	0%	71.46	0	0	0%
21	492.79	13	14.25%/2792/2851	61.64%	447.97	5	3.47%/5893/5731	7.03%
22	339.83	0	0	27.87%	310.79	0	0	0.96%
23	271.17	0	0	0.06%	98.31	0	0	0.07%
24	123.67	0	0	0%	72.49	0	0	0%

(a/b/c): a* is the average optimality gap; *b* refers to the average best solution; *c* denotes the average optimal solution. All those values just concern the cases whose computation times are more than 600 s.

We also tested the FCFS benchmark and the THO model. In those cases, train delays in the optimal solutions considering objective 1 can be reduced by 22% to 35%, and by 17% to 24%, for FCFS and THO respectively (see Fig. 21). For FCFS, in comparison with the cases related to heterogeneous speeds timetable, the current case of timetable with homogeneous speeds leads to smaller differences and relatively worse performances. Instead, the gap between the solution of THO and the proposed model increases, probably due to more interactions between trains and the generally higher train speed.

Table 8 reports comprehensively computation times and optimality gaps of both objectives (objective 1 on columns 2–5; objective 2 on columns 6–9). The first column is the index of cases, as in Table 7. We consider a maximum computation time (time limit) of 600 s. The average computation times over the 20 primary delays cases are shown on the second and sixth columns; actually, the vast majority of the cases find the optimal solution within 600 s. The amount of cases hitting the time limit is shown on the third and seventh column. Column 4 (respectively 8) reports useful information for those

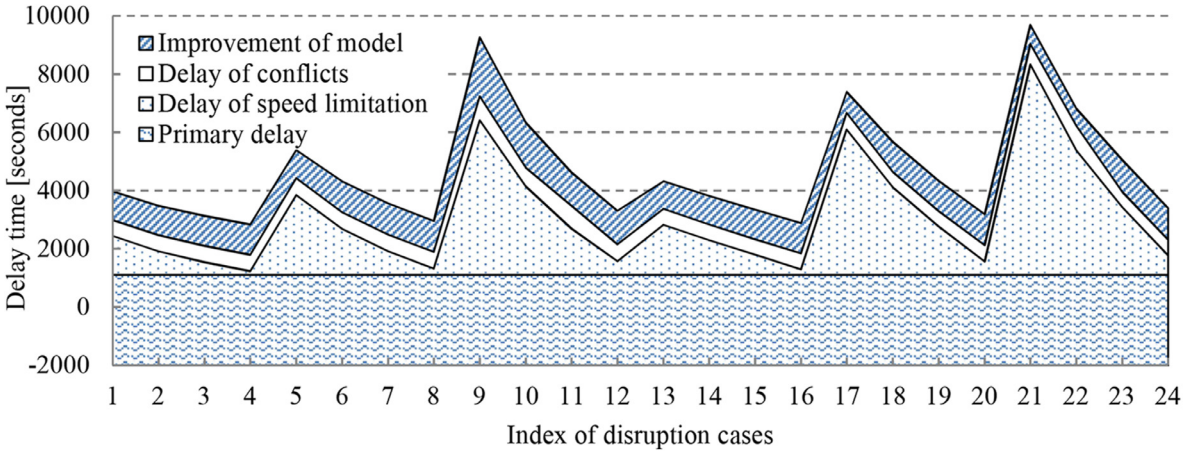


Fig. 22. Delay components of objective 1 under speed limitation.

cases hitting the time limits of 600 s, namely average optimality gap (a), average best solution (b) reported. Furthermore, by extending the computation time to 2.5 h for those cases, all the optimal solutions (c) are obtained; the average of those optimal solutions are listed in column 4. Columns 5 and 9 report the optimality gap for a shorter computation time limit of three minutes. In term of the average computation time, most cases manage to get the optimal solution within 3 minutes, and only few runs go beyond 5 minutes. In general, the optimization process for objective 1 is slower than that for objective 2. In any case at 10 minutes, no less than 75% of the cases solved by objective 1 can reach the optimal solution, while for objective 2, the rate goes even higher, to 83%. In general, the more serious the disruption is, the more computation time is needed. Regarding the gap (a) between the best solution and optimal solution in column 4 or 8, although the optimal gaps (a) in some cases are quite large, the differences between best objective values (b) and optimal objective values (c) vary from 0 s to 177 s and the average difference is 69 s. In other words, the objective value of the optimal solution is only around 1 minute smaller than that of the best solution. This shows that the best solution within the given time limitation of 10 minutes is very similar with the optimal solution which could be found only with a much longer computation time. As shown in columns 5 and 9, the solving method is able to get a solution with a gap of no more than 5% for 75% cases. Sometimes, the solution with the computation time of 3 minutes looks directly acceptable for what concerns quality. The amount of cases whose gaps exceed 10% is rare, with just 6 or 5 cases (for objective 1 and objective 2 respectively); clearly this percentage is relatively larger than that with 10 minutes time limits. Thus our approach is able to get a solution with a good quality, and a generally small optimality gap. We believe this is due to the good upper bound generated by the two-step procedure, which is already close to the optimal value in an earlier stage, while the lower bound computed in CPLEX improves very slowly or not at all.

To summarise, Fig. 22 reports the available improvement factors and gaps for the 24 disruption scenarios, objective 1, similarly to Fig. 12. The layer on the bottom (with waves) is the delay resulting from primary delays (6221 s), combined with about 2000 s of recovery time, which result in roughly 1000 s of final delay. This is in line with the situation in the previous test cases, and with a larger recovery time. The layer above (dotted) is the delay caused by speed limitation. The value of this part depends on the duration and severity of speed limitation. It is clear that the severity of speed limitation plays the key role in introducing delay; but with the same severity of disruption, the homogeneous-speed timetable results in less delay than the heterogeneous-speed timetable of Section 5.1. The layer immediately above (white) denotes the delay resulting from the unavoidable conflicts. On average, this accounts for about 500 s. This is much smaller than the one found in the heterogeneous-speed timetable, due to a higher regularity of the traffic running. The top layer refers to the gap between FSFS and the proposed model (i.e., the improvement of the model), which is around 1000 s on average and reaches up to 2000 s in this case. The conflict resolution computed by the proposed model is able to reduce about two thirds of secondary delays caused by the conflicts under the combined influence of speed limitation and primary delay, for this particular timetable with homogeneous speed.

6. Conclusions and future research

We solve the real-time train rescheduling problem to minimize secondary delay for high-speed railway traffic with a quasi-moving block signalling system. Disruptions causing dynamic, varying speed limitations and primary delays are considered during the optimization process. A mixed integer linear programming model is proposed which integrates speed management principles (choosing the operating speed level of trains on block sections) and traffic management principles (determining times and orders of trains), to coordinate dynamically the distance headway between trains according to the train speed and the quasi-moving block operational rules. A good solution can be found by the proposed model within the

computation time of 10 minutes on a commercial solver, by employing a two-step search procedure. Two comprehensive sets of computational experiments are performed, in a realistic test case with heterogeneous speed traffic, and on a part of the Wuhan-Guangzhou high-speed railway with homogenous speed traffic.

According to the analysis of the experimental tests, we draw the following conclusions:

(1) An increasing amount of sections disrupted results in a corresponding linear increase of delays. The relation between duration of disruption (maximum permitted speed level) and secondary delay leads to more complex effects, which differ further in the cases of the two objective functions considered (objective 1: minimization of secondary delay; objective 2: minimization of positive secondary delay). The lightest disruptions can be totally absorbed by the timetable by means of objective 1.
(2) Within the gap defined by a benchmark upper bound given by the FSFS rule, and a benchmark lower bound obtained by relaxing the capacity constraints, the optimized solution achieves a delay reduction compared to the original timetable by 70% and 90%, for objective 1 and objective 2 respectively. The optimization is able to reduce delay times due to conflicts by half and two thirds, for the cases with heterogeneous and homogenous traffic, respectively. For the real-life test case, we solve 92% (respectively 97%) of the instances to optimality within few minutes of computation, respectively for objective 1 or 2.
(3) Minimizing only the positive delay (objective 2) results typically in solutions with less quality, but a faster solution compared to objective 1, which allows for discounting positive delays with negative secondary delays.
(4) The dynamic rule FCFS improves the solution compared to the offline static rule FSFS keeping the timetable order, but results in delay times 20% larger compared to the solution of the proposed model.
(5) The straightforward extension of existing fixed block signalling models, which is a time-headway method based on Time Headway Only (THO), performs worse than our approach. In fact, compared to the proposed model including both time and distance-based headway, there are always positive gaps which can reach up to 400 s. Our approach shows a larger gap of the improved performance for the homogeneous timetable, compared to the solution of the THO model.

Moreover, this paper also proves the benefit of real-time rescheduling for quasi-moving block signalling systems, which have been so far neglected by academic research. For those systems, intelligent speed management can decrease the amount of conflicts and increase fluidity of traffic. Changing train orders is particularly effective in the case of heterogeneous timetables, where fast trains can avoid the hinder from slower traffic and reach their planned speed faster after a speed limitation.

By adjusting the parameters related to speed level, additional running time and length of the section, based on the performance of acceleration or deceleration of the trains, it is possible to generalize the method for conventional railway lines or other quasi-moving block systems. Moreover, the inclusion of speed in the optimization problem could allow for integrated speed management and traffic control optimization. Future research directions opened up by the present research include a more detailed characterization of uncertainty of a single, or multiple disruptions. Under the given disruptions and delays, we can investigate their influence on train operations and give robust advices for high-speed traffic, reducing extra travel time which jeopardizes the concept of short travel time related to high-speed traffic. Further developments should also focus on more efficient algorithms to deal with larger and more complex instances, possibly scaling up to the level of a network. One possible direction is to consider rolling horizon schemes, which update control actions as more information on the disruption is available (see Samà et al., 2017b for a similar approach). Otherwise, a set of feasible and potentially interesting speed profile options could be generated by means of column generation or Lagrangian relaxation, then the problem would be to assign one such profile for each train. Algorithmically, it will be worthwhile to extend the degrees of freedom of the model by considering more fine-grained speed profiles, or rerouting. Those two directions require the solution of a substantially increased combinatorial complexity.

Acknowledgments

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Appendix A. Analysis of traffic for objective 1 and objective 2

We investigate the effect of the solution with regard to the detailed traffic, by focusing on a single disruption, namely $<a,b,c> = <0.5\,\text{h},\ 1,\ 50>$, and analysing the distribution of final delays with 20 delayed cases. Thus the reliability of the resolutions is assessed, i.e., solutions with less delay and less variation of delays. Fig. 23 reports distributions of the final delays (i.e., the measured deviation between actual arrival and planned arrival) related to 20 trains (x-axis), in terms of the 95% confidence range. For each train, three bars are plotted, in sequence blue (Optimizing objective1), red (Optimizing objective2) and green (FSFS solution).

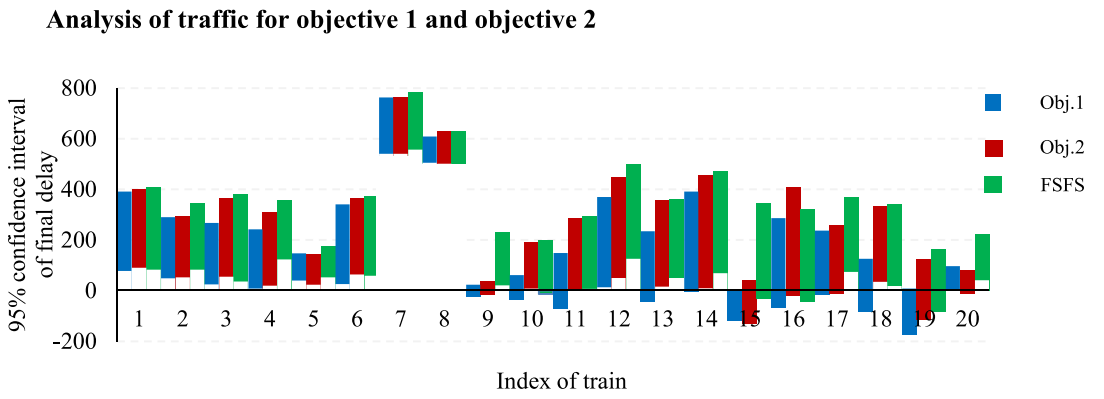


Fig. 23. 95% confidence interval of final delay for each train, disruption ⟨0.5 h, 1, 50⟩. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

Consistently, the bottoms of the blue bars are lower than that of red bars, reflecting that the objective 1 reduces more delays for trains; the small gap gives an idea of the extent by which Objective 1 exploits early arrivals. The blue bars are sometimes negative, i.e., the train is actually arriving ahead of schedule. As a result, objective 1 delivers better solutions, as early trains can be exploited to reduce the overall delays. There are also a few trains (7 and 8 for instance), which are systematically delayed, as they are affected the most by this specific disruption considered. Much delay cannot be reduced, especially when it is depending on primary delay and speed limitation. In terms of the lengths of the bars, i.e., the dispersion of the samples, which describe the reliabilities of solutions, objective 1 and objective 2 achieve more compact dispersions than that of FSFS. This is particularly evident for some trains like 15 and 20. Overall, objective 1 can achieve consistently good solutions when large primary delays are present, and disruptions are not particularly strong, otherwise objective 2 is a more effective choice. In practice, the two optimization processes could be performed in parallel and presented to the traffic controllers to choose one particular solution out of the two.

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